AROMATIC COMPOUNDS

(I) Aromaticity (In GOC I)

(II) Electrophilic Sub. Rxn (ORM-II)

(III) Benzene

(III.A.) Nitrobenzene
(III.B.) Chlorobenzene

(IV) Benzene diazonium salt

(Covered in ORM and Reduction-Oxidation)

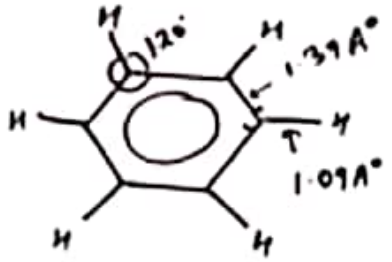
(V) Phenol

(VI) Aniline / Amine

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AROMATIC COMPOUNDS

(III) Benzene:-



$6\pi e^-$

- MP ≃ 5°c
- BP ≃ 80°c

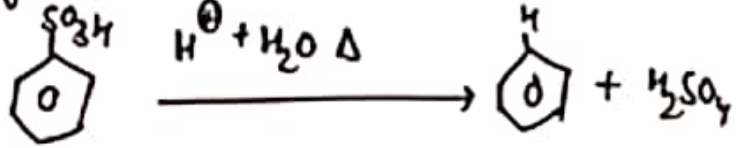
Soluble in CCl_4, $CHCl_3$, Acetone etc

Formation of Benzene:-

(1) By Grignard reagent:- (ORM-V)

$$Ph\text{-}MgBr + H_2O \xrightarrow[Acid]{(H\text{-}2)} Ph\text{-}H + Mg^{Br}_{OH}$$

(2) By desulphonation of Benzene sulphonic acid (ORM-II)



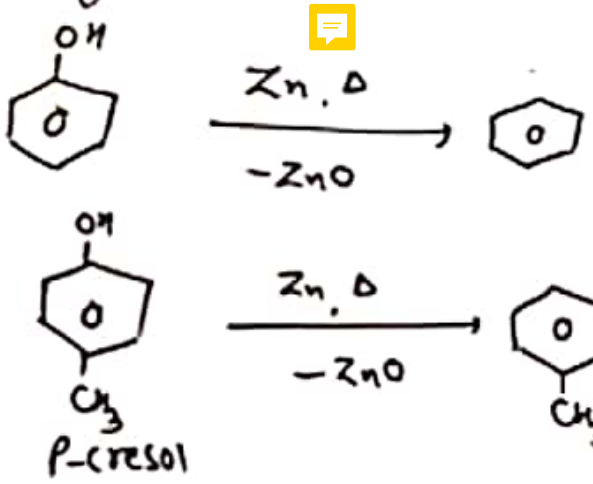
Scanned with CamScanner

AROMATIC COMPOUNDS

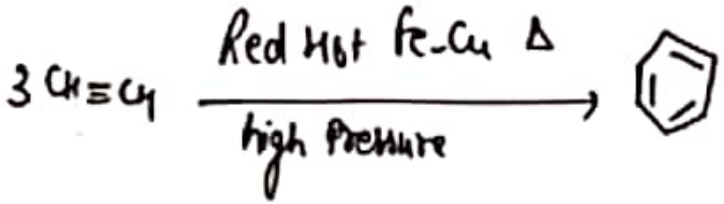
(III) Benzene:-

Formation

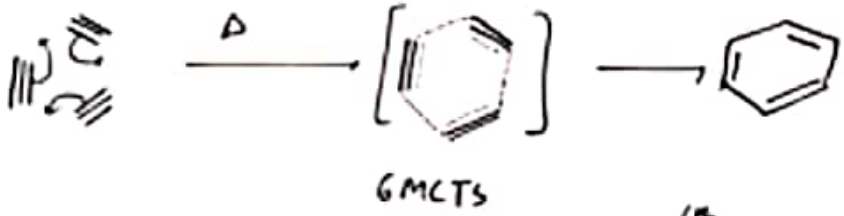
③ By Phenol



(b) By Tricyclisation of acetylene:-



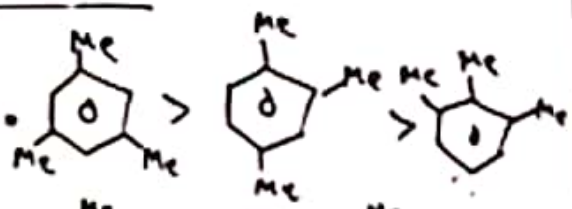
mechanism



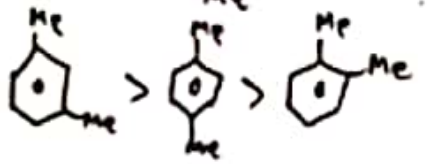
eg: $3\ CH_3-C\equiv CH \xrightarrow[\text{Fe-Cu } \Delta]{\text{Red Hot}}$ mesitylene

Thermodynamic Stability

» Trimethyl benzene



» Dimethyl benzene »

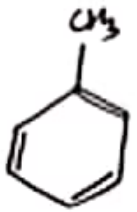


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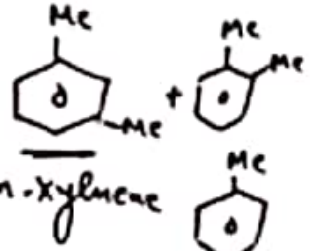
AROMATIC COMPOUNDS

(III) Benzene:-

Formation

eg3 $H-C\equiv C-H + CH_3-C\equiv CH \xrightarrow{Fe,\ \Delta}$  Toluene

2 : 1

eg4 $H-C\equiv C-H + CH_3-C\equiv CH \longrightarrow$  m-Xylene + 

1 : 2

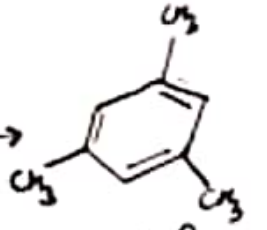
eg5 $H-C\equiv C-H + H-C\equiv N \longrightarrow$  Pyridene.

2 : 1

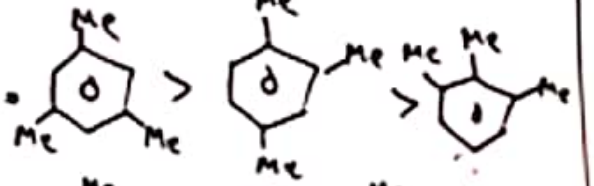
mechanism

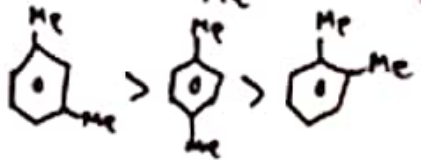


GMCTs

eg1 $3\ CH_3-C\equiv CH \xrightarrow[Fe-Cu\ \Delta\ Pressure]{Red\text{-}Hot}$  Mesitylene

Thermodynamic Stability

⇒ Trimethyl benzene ⇒ 

⇒ Dimethyl benzene ⇒ 

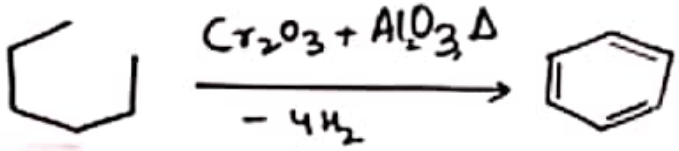
Scanned with CamScanner

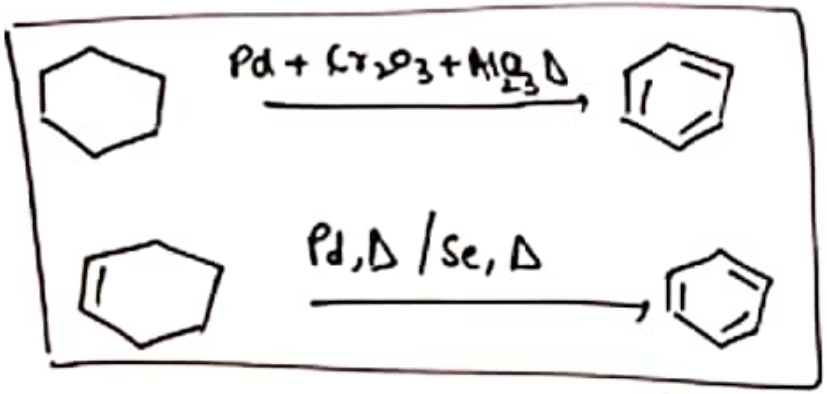
AROMATIC COMPOUNDS

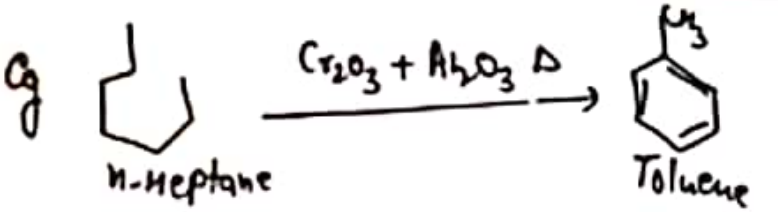
(III) Benzene:-

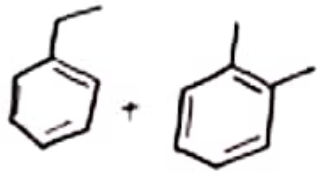
Formation

⑤ By aromatisation of n-Hexane:-







eg n-octane → 

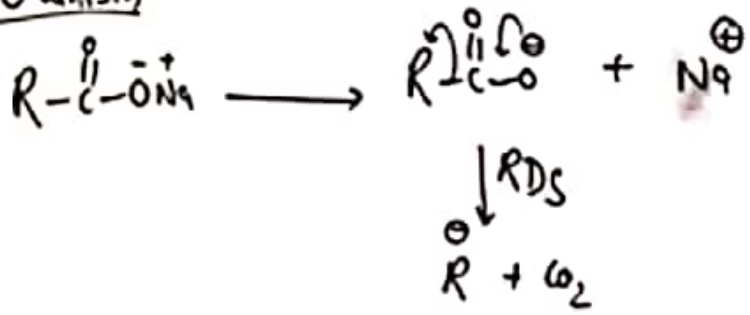
(6) By decarboxylation of benzoic acid :-

$$R\text{-}COOH/R\text{-}COONa_{aq} \xrightarrow[\text{Soda lime}]{(NaOH + CaO)\ \Delta} R\text{-}H + Na_2CO_3$$

degradation method:
to reduce no. of C

$$Ph\text{-}COONa \xrightarrow{NaOH + CaO\ \Delta} Ph\text{-}H$$

mechanism



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AROMATIC COMPOUNDS

(III) Benzene:-

Formation

$$R^{\ominus} + H_2O \longrightarrow R\text{-}H + \overset{\ominus}{O}H$$

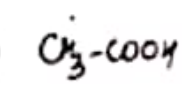
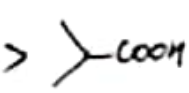
Carbanion

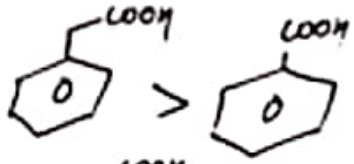
$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

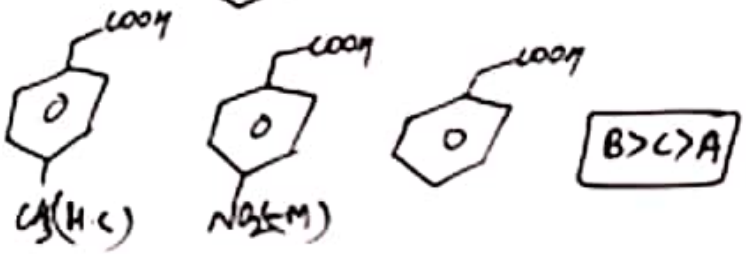
⇒ Intermediate = carbanion

→ Reactivity of decarboxylation ∝ Stability of anion inter... ($1^\circ > 2^\circ > 3^\circ$)

Q Give Reactivity order for decarboxylation.

① $CH_3\text{-}COOH > \text{  } > \text{  }$

② 

③ 

④ 

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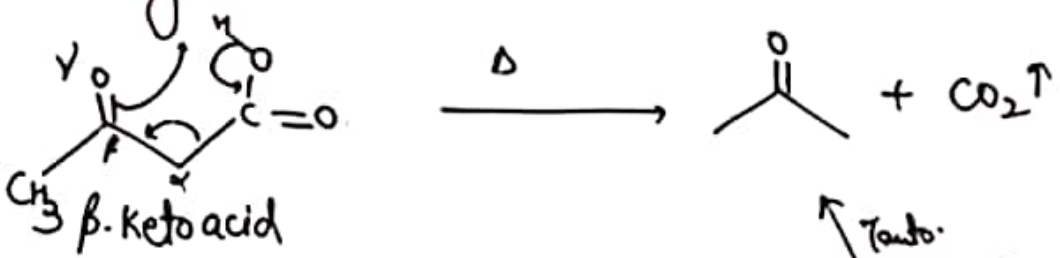
AROMATIC COMPOUNDS

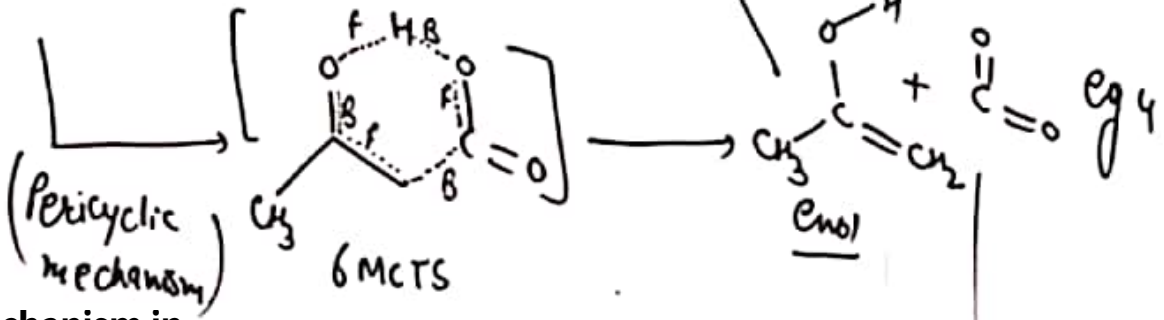
(III) Benzene:-

Note When a strong -M group present on α Position

(-M – $\overset{\alpha}{CH_2}$ – COOH) of –COOH group,

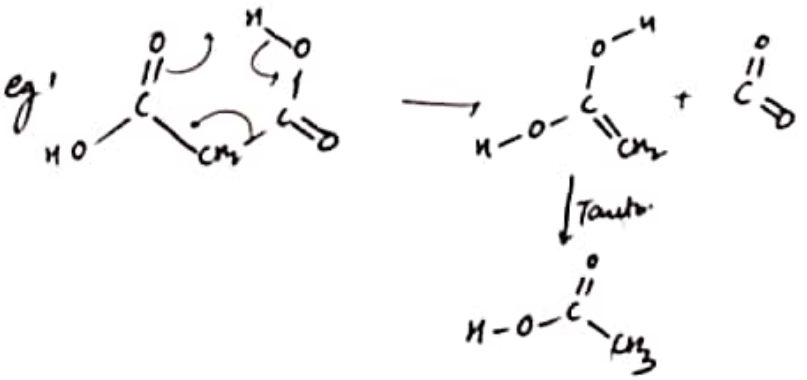
then decarboxylation possible only on high temp without the need for soda-lime





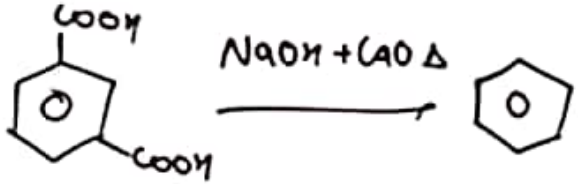
mechanism in which cyclic TS forms

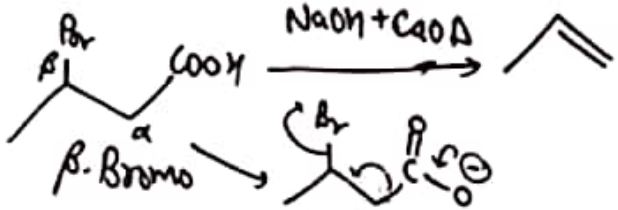
eg 1



eg 2 $CH_3-COOH \xrightarrow{NaOH + CaO, \Delta} CH_4$

eg 3



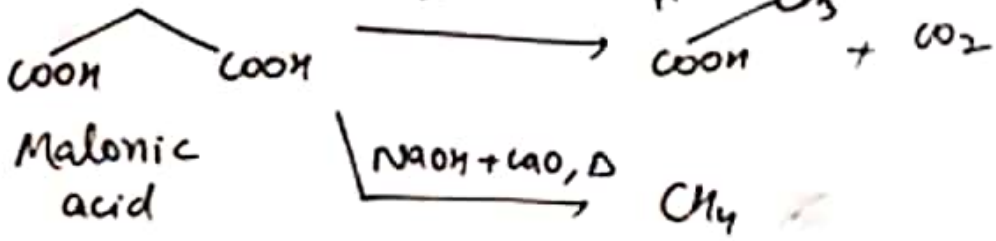


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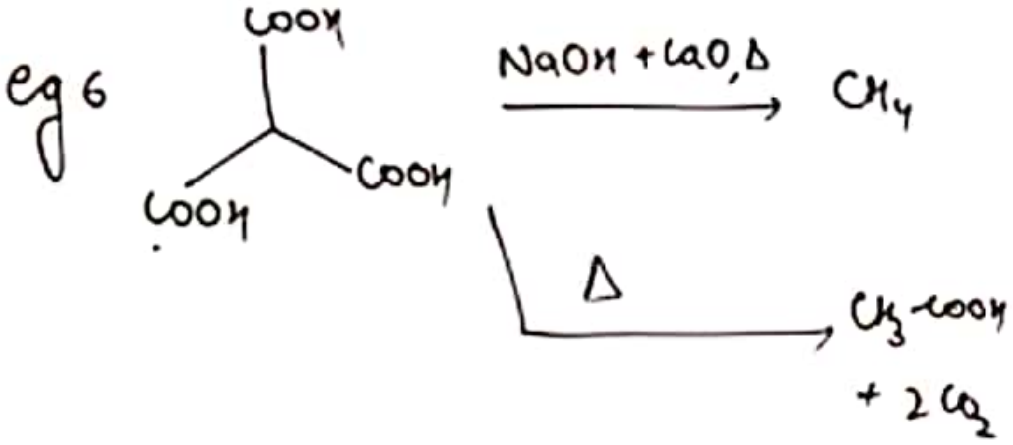
AROMATIC COMPOUNDS

(III) Benzene:-

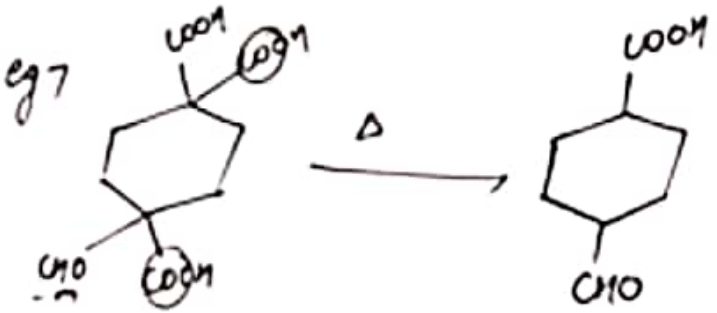
eg 5



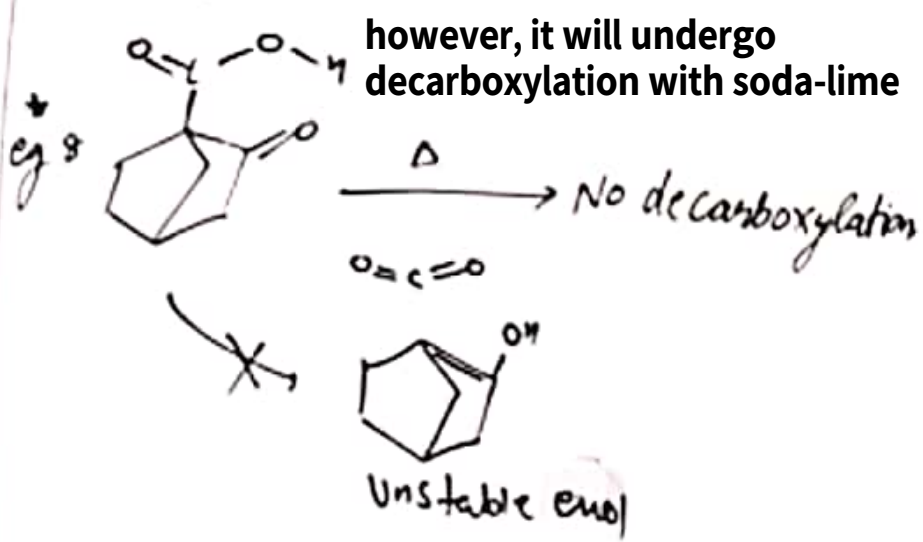
eg 6



eg 7



eg 8



however, it will undergo decarboxylation with soda-lime

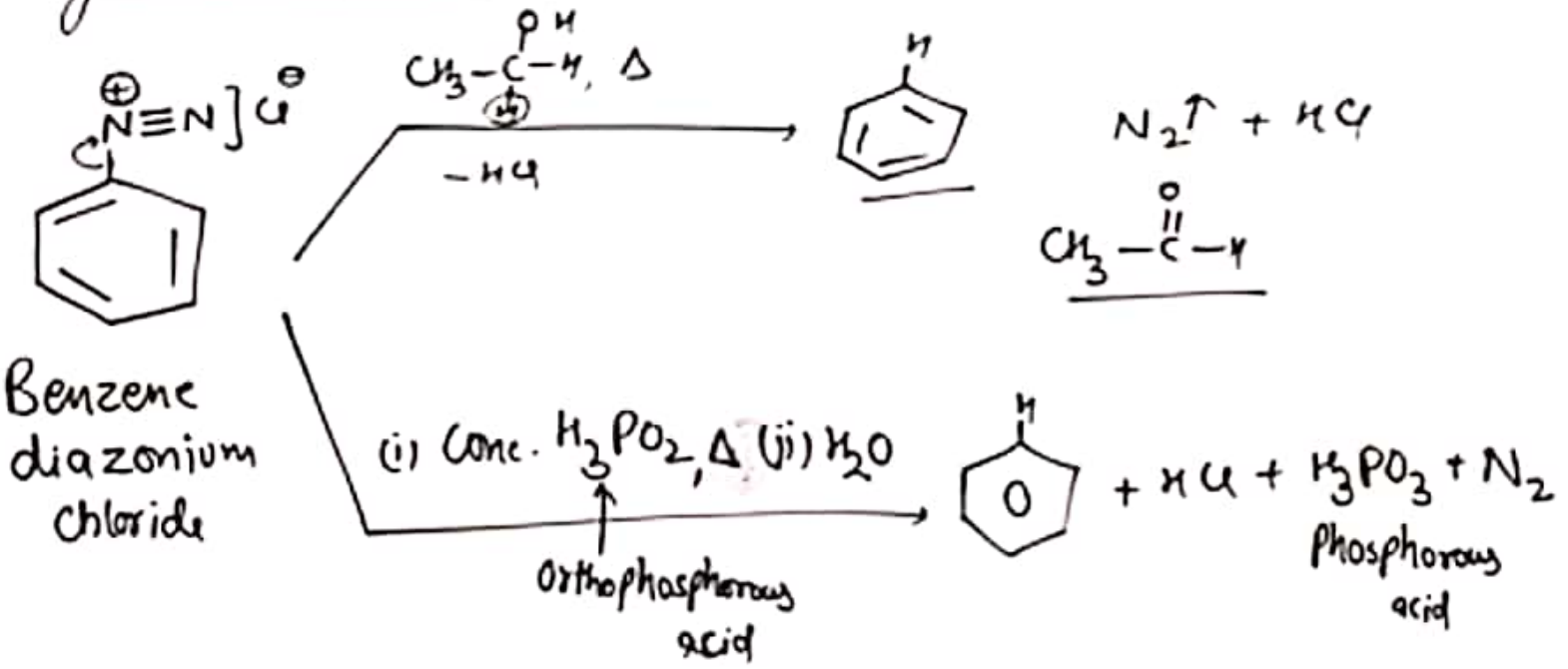
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AROMATIC COMPOUNDS

(III) Benzene:-

Formation

(1) By Benzene diazonium salt (Imp.)



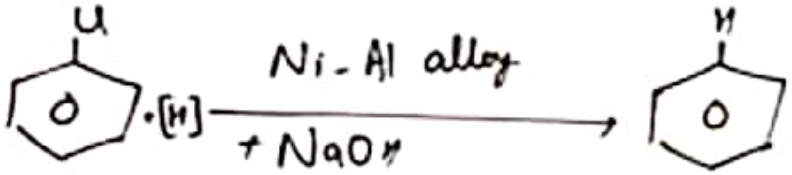
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AROMATIC COMPOUNDS

(III) Benzene:-

formation:-

(8) From chloro benzene:-



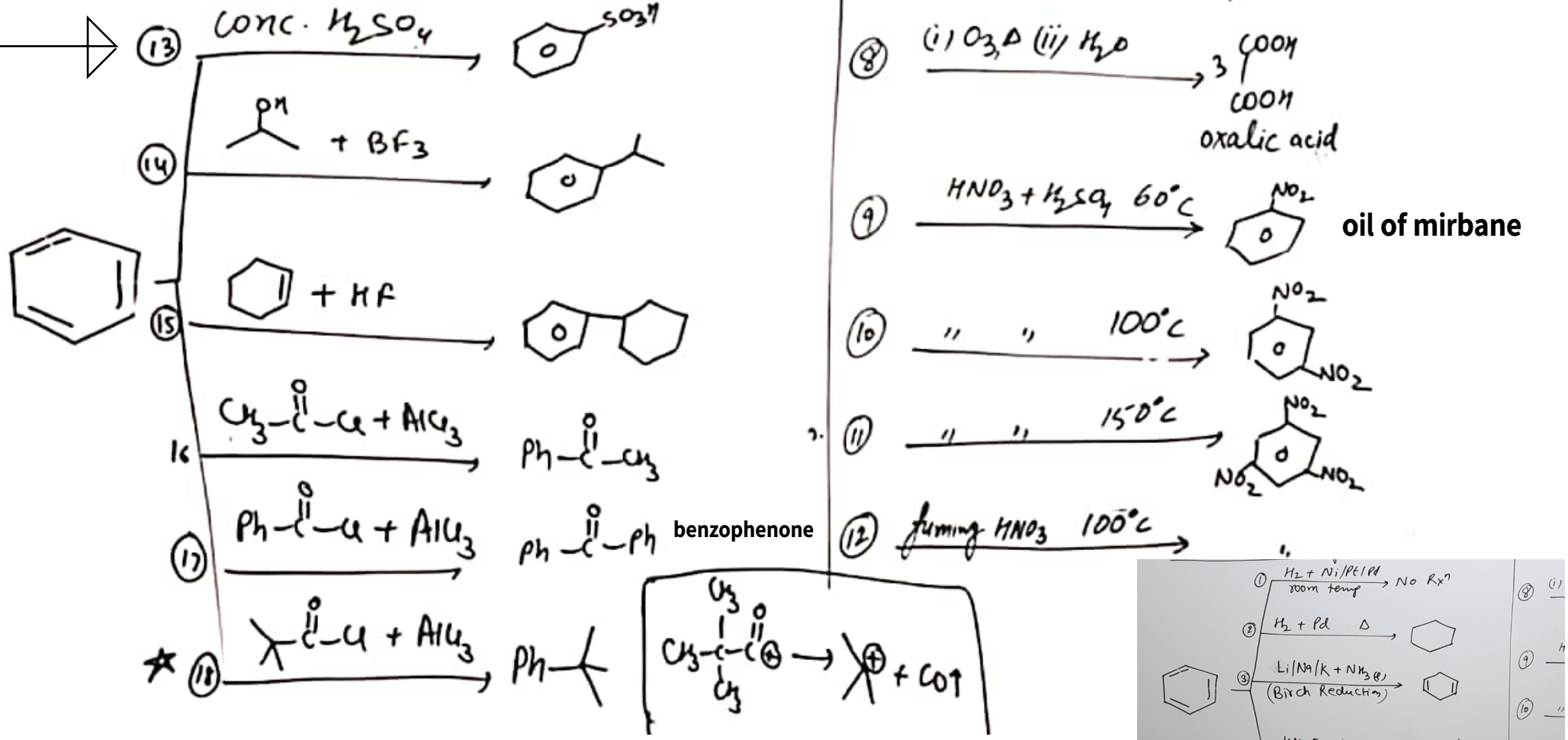
+ AlCl3
+ sodium aluminate

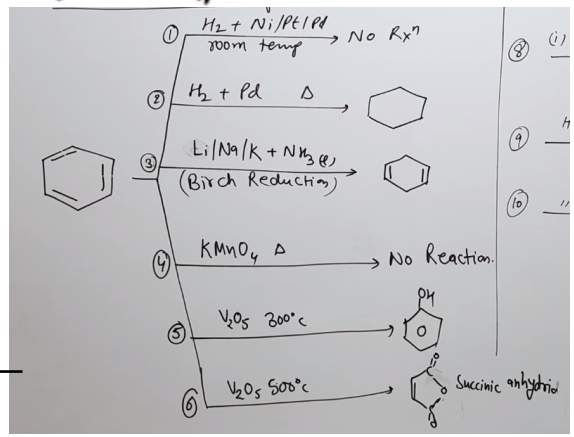
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AROMATIC COMPOUNDS

(III) Benzene:-

Chemical Reactions of Benzene:-

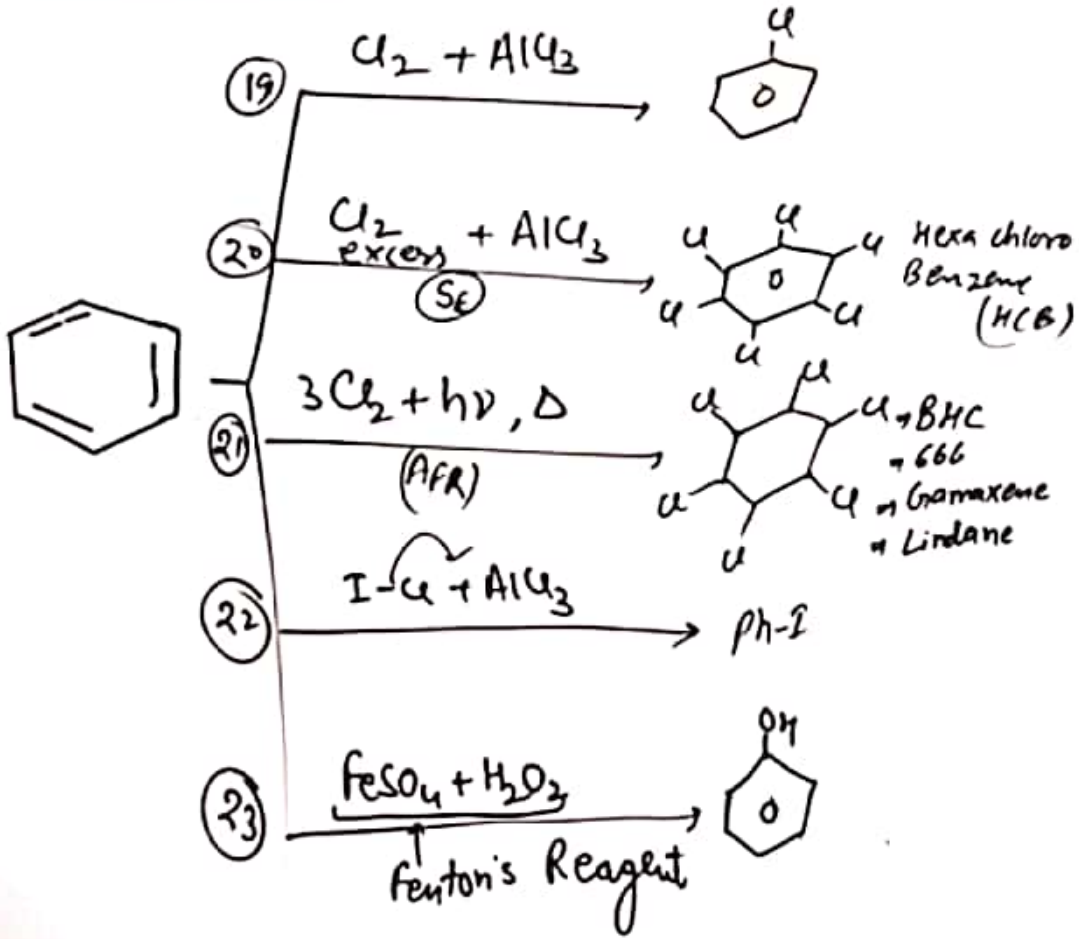




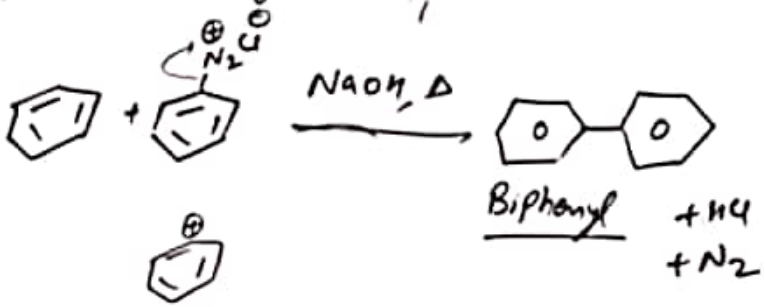
AROMATIC COMPOUNDS

(III) Benzene:-

Chemical Reactions of Benzene:-

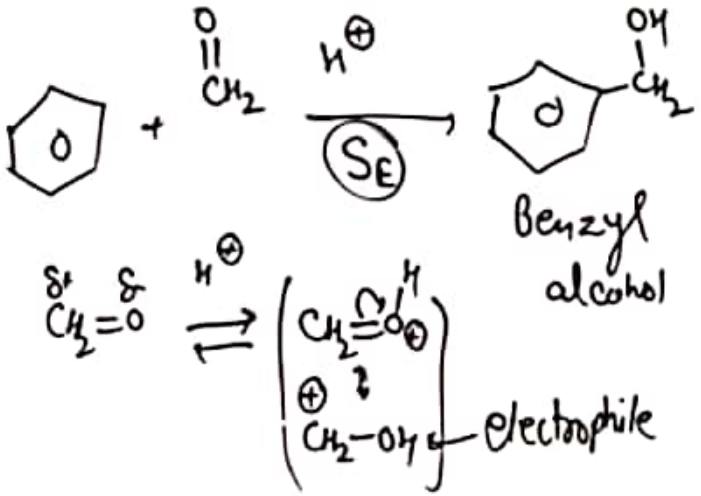


24 Gomberg's Reaction



25 Reaction with aldehyde:-

rex. is difficult with ketone



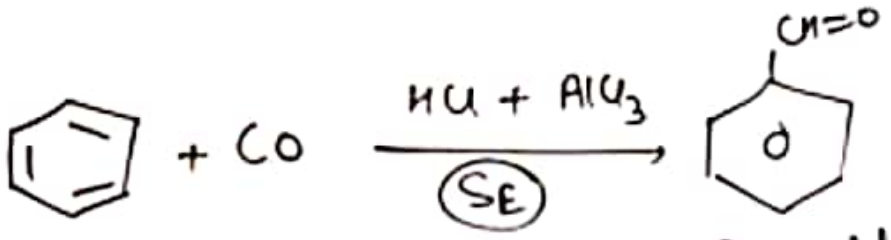
Scanned with CamScanner

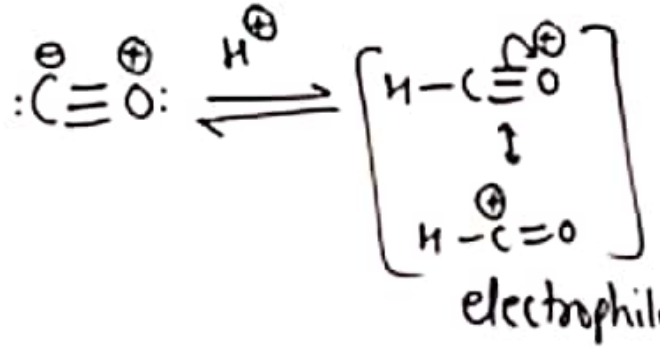
AROMATIC COMPOUNDS

(III) Benzene:-

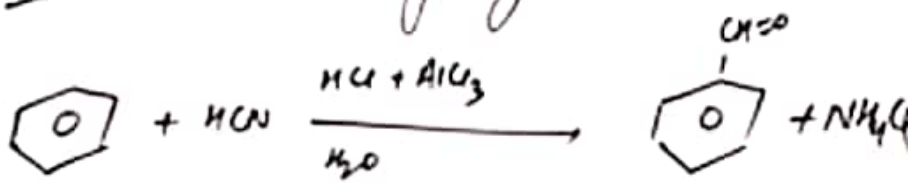
Chemical Reactions of Benzene:-

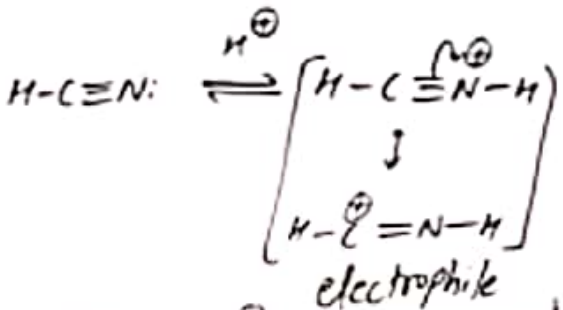
26 Gattermann's Koche formylation:-

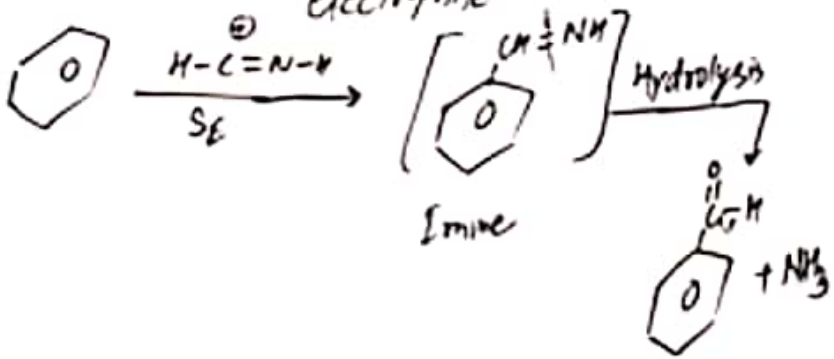




27 Gattermann's formylation







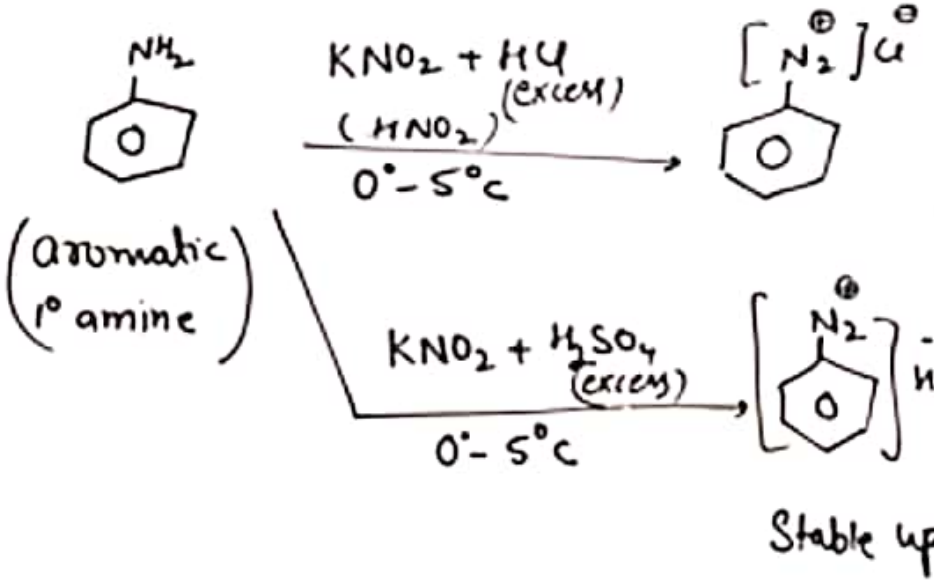
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AROMATIC COMPOUNDS

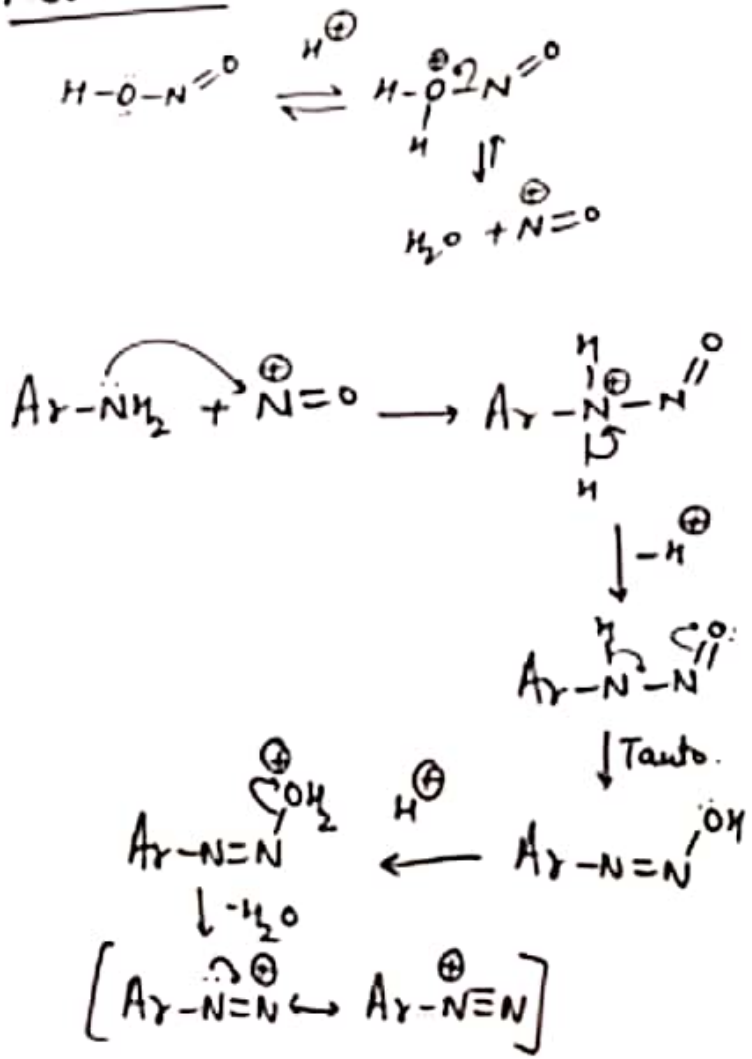
(IV) Benzene diazonium salt:-

Formation

Diazotisation



Mechanism

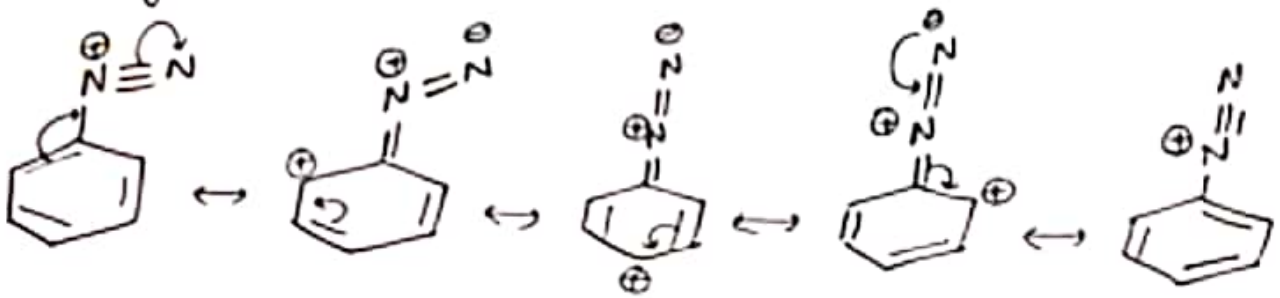


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AROMATIC COMPOUNDS

(IV) Benzene diazonium salt:-

Resonating structure of diazoniom salt:-



For Aliphatic 1° amine:-

$$R-NH_2 \xrightarrow[0^\circ-5^\circ C]{KNO_2 + HCl} R-OH + R-O-R + Alkene..$$

$$R-NH_2 \longrightarrow R-\overset{\oplus}{N_2} \longrightarrow R^{\oplus} \xrightarrow{SN^1 / H_2O} + N_2 \uparrow$$

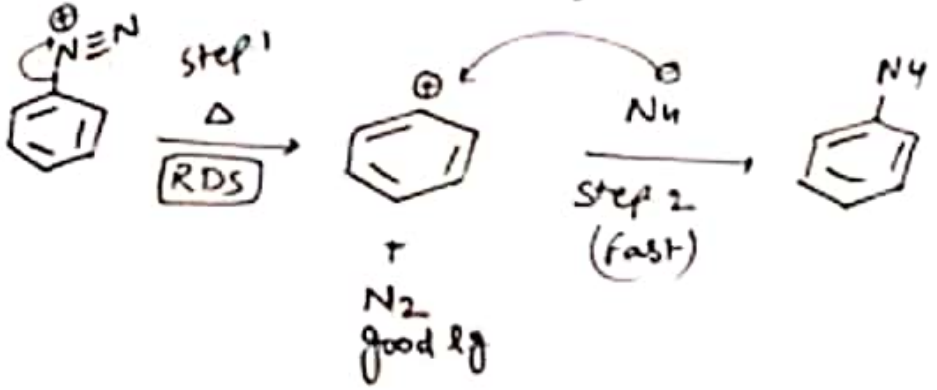
unstable

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AROMATIC COMPOUNDS

(IV) Benzene diazonium salt:-

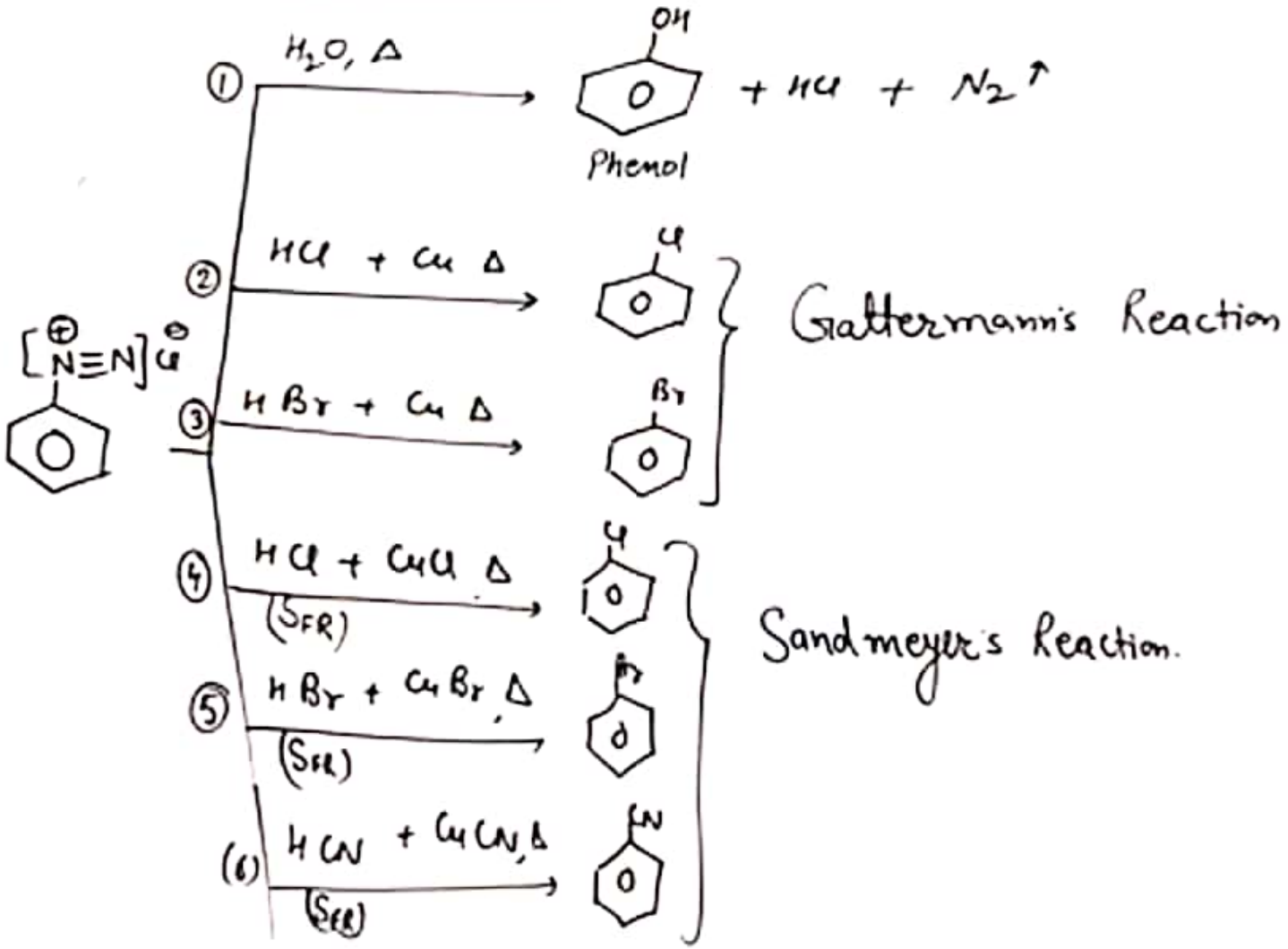
Chemical Reaction of diazonium salt :- mainly gives ArSN1 Reactions.



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AROMATIC COMPOUNDS

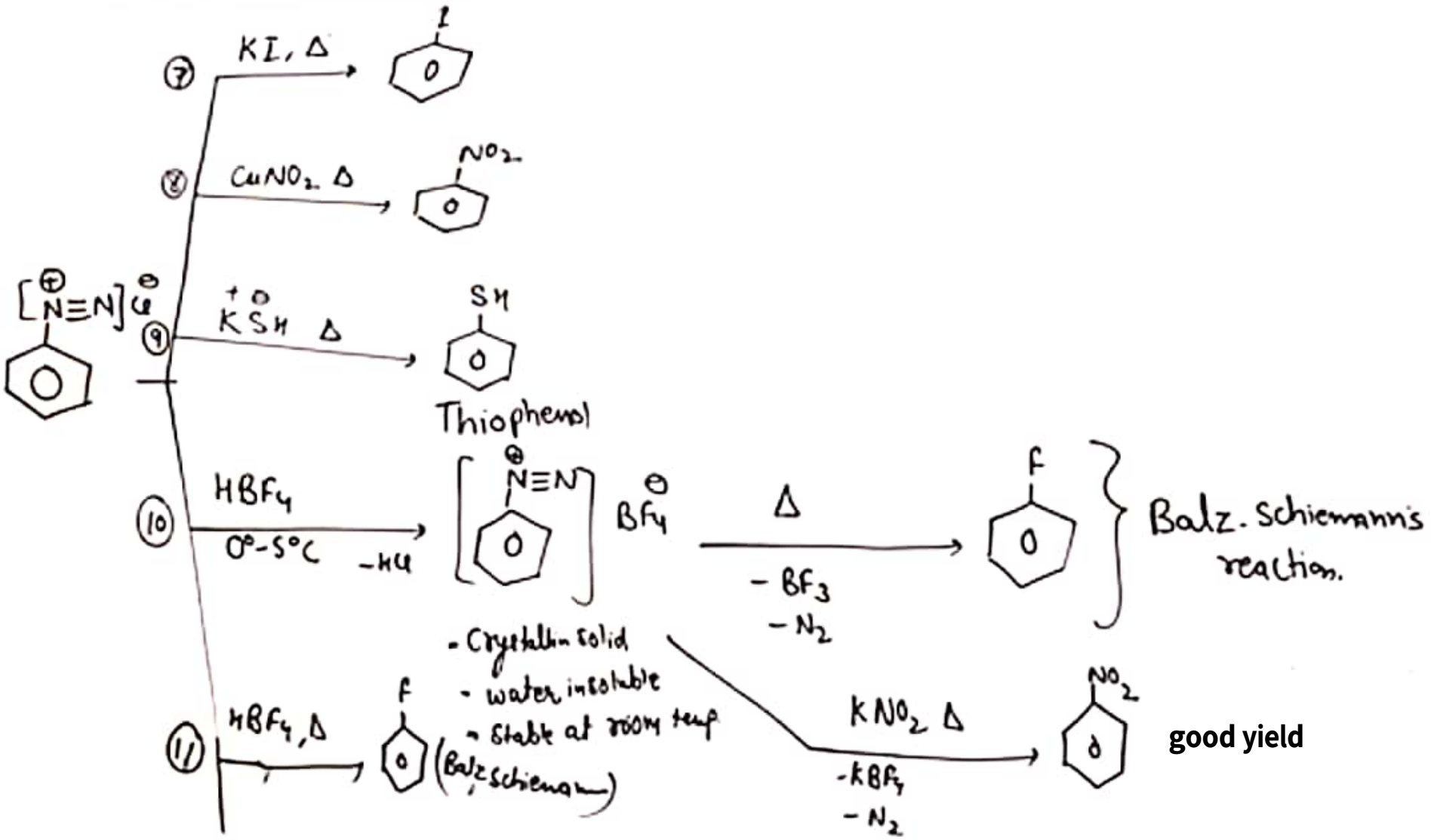
(IV) Benzene diazonium salt:-



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AROMATIC COMPOUNDS

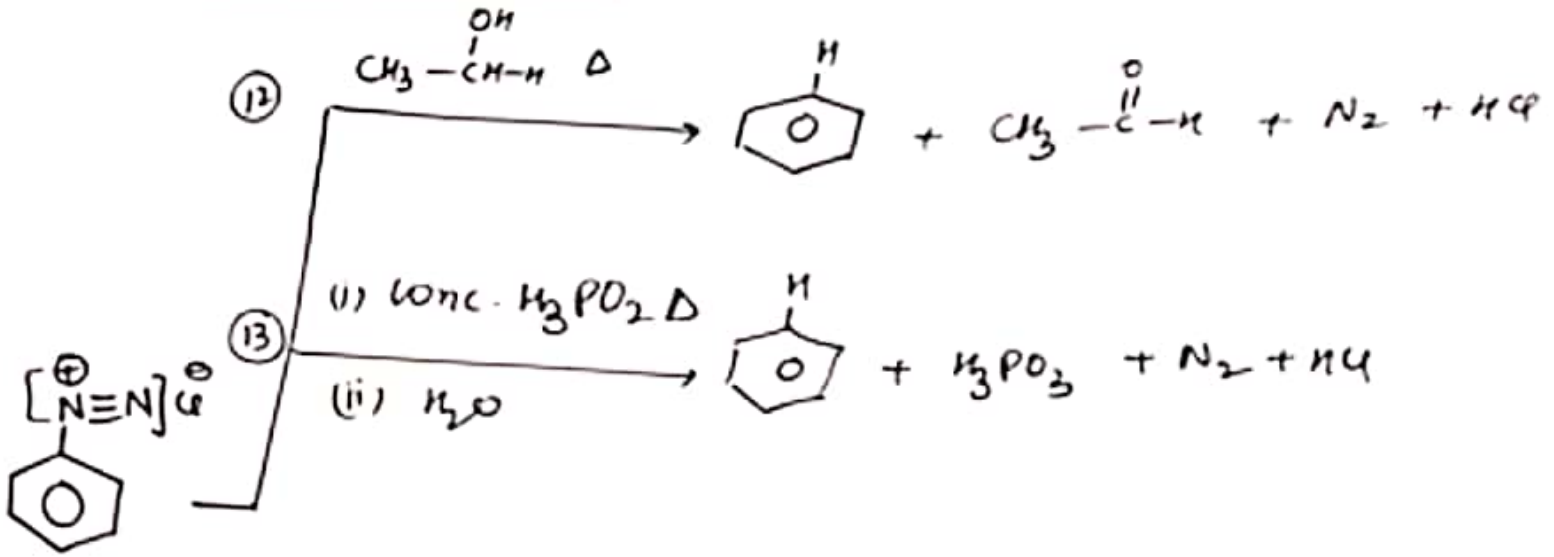
(IV) Benzene diazonium salt:-



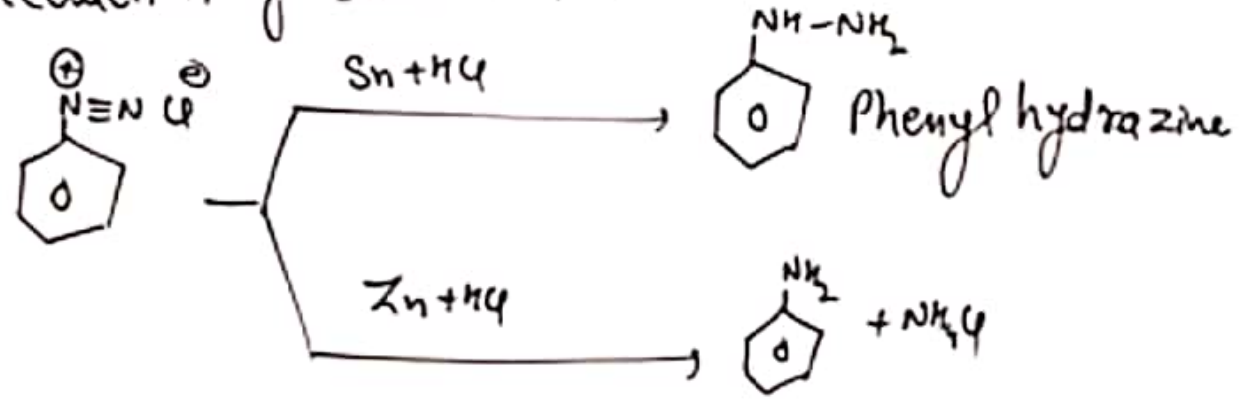
Scanned with CamScanner

AROMATIC COMPOUNDS

(IV) Benzene diazonium salt:-



(14) Reduction of Diazonium salt:

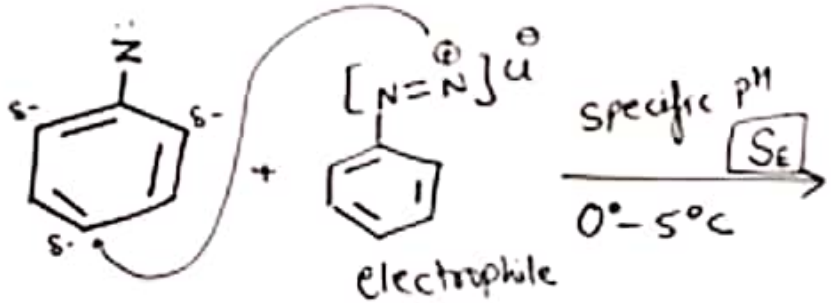


Scanned with CamScanner

AROMATIC COMPOUNDS

(IV) Benzene diazonium salt:-

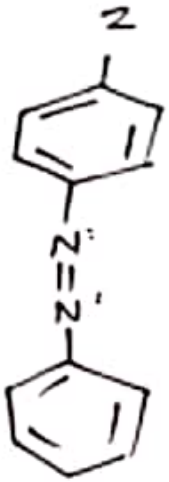
(15) Coupling Reaction:-



specific pH

S_E

0°-5°C



Azo dye

(Syn + Anti)

Geo. isomers

⇒ only para product (Steric hindrance)

⇒ Specific colour (due to conjugation)

ν high

$Z = -OH, -NH_2, -NHR, -NR_2$ etc

Basic: pH = 8-9

acidic: pH = 5-6

if para is blocked, then rex. takes place at ortho

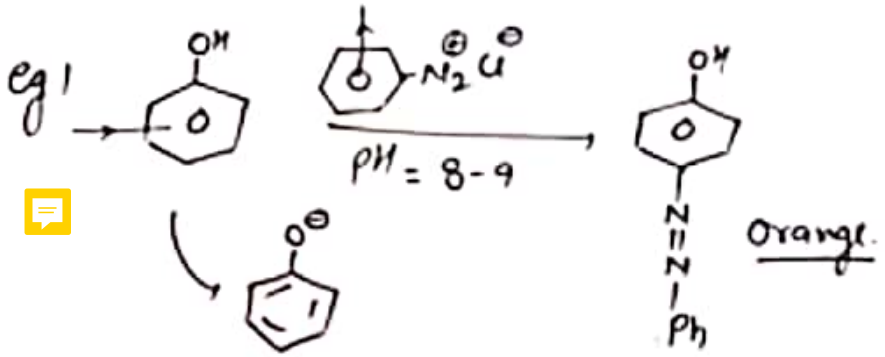
Scanned with CamScanner

AROMATIC COMPOUNDS

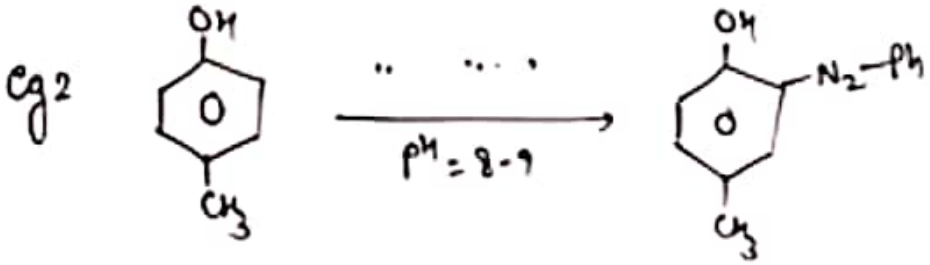
(IV) Benzene diazonium salt:-

(15) Coupling Reaction:-

eg 1



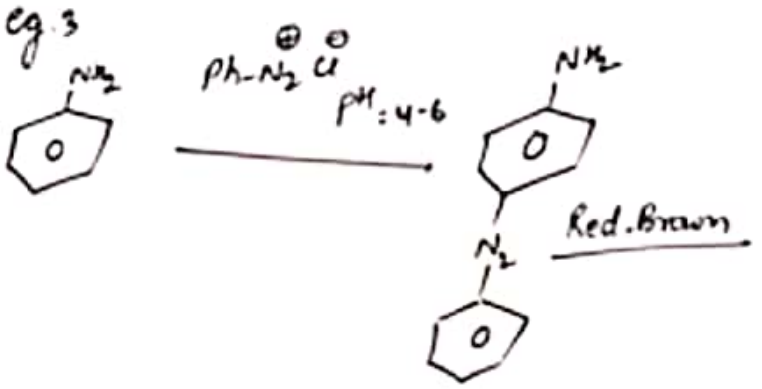
eg 2



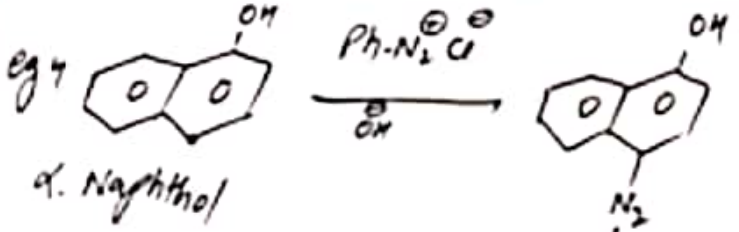
$Ph\text{-}N_2Cl \xrightarrow[pH>10]{OH^\ominus} Ph\text{-}N=N\text{-}OH + Cl^\ominus$

No coupling Rxn

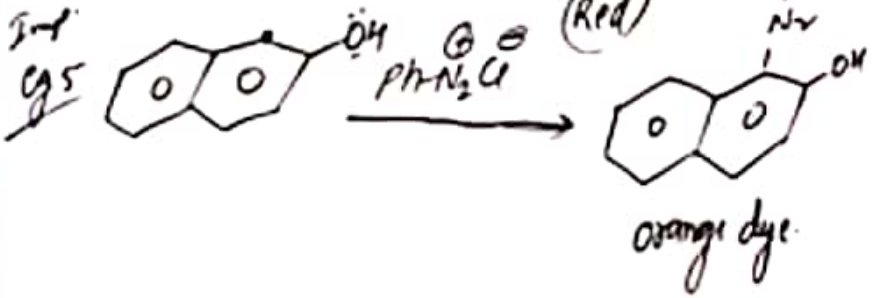
eg 3



eg 4



eg 5



Q. Why no coupling takes place at pH>10 (i.e highly basic solution) ?

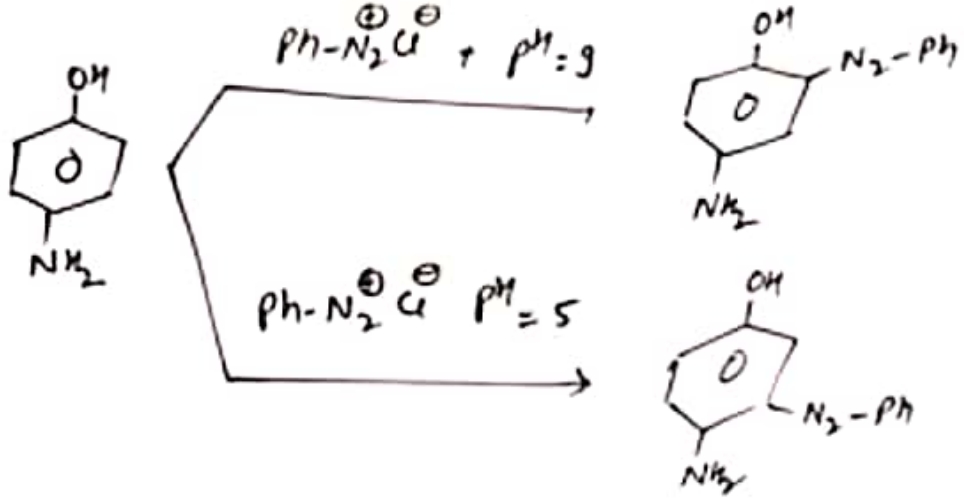
Scanned with CamScanner

AROMATIC COMPOUNDS

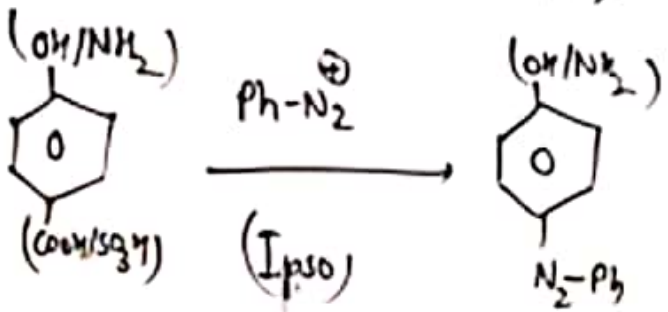
(IV) Benzene diazonium salt:-

(15) Coupling Reaction:-

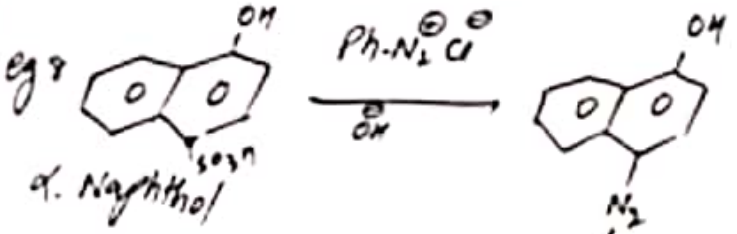
eg 6



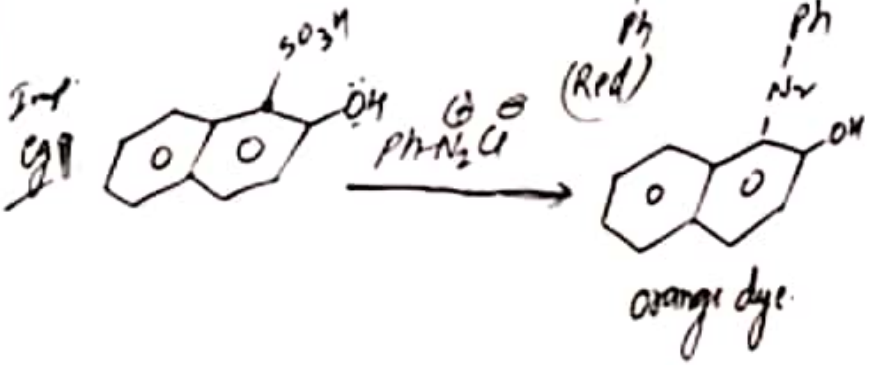
eg 7



eg 8



eg 9



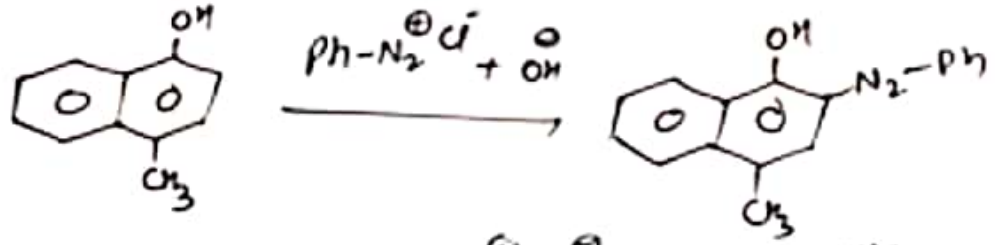
Scanned with CamScanner

AROMATIC COMPOUNDS

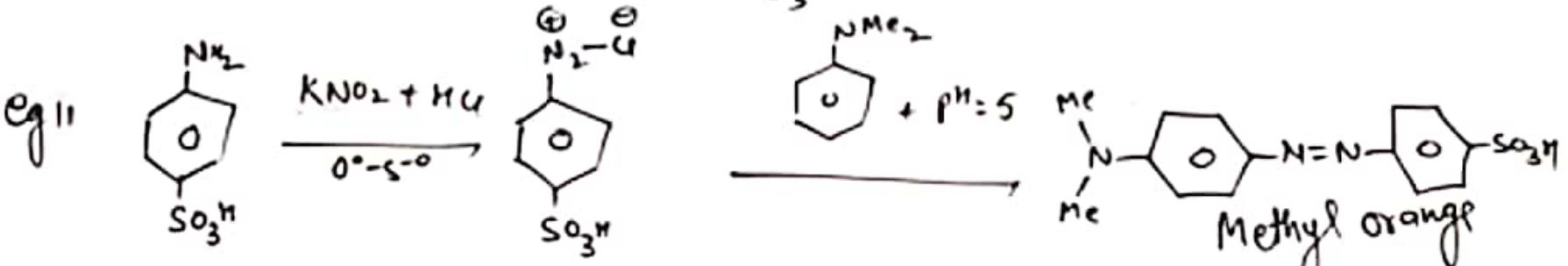
(IV) Benzene diazonium salt:-

(15) Coupling Reaction:-

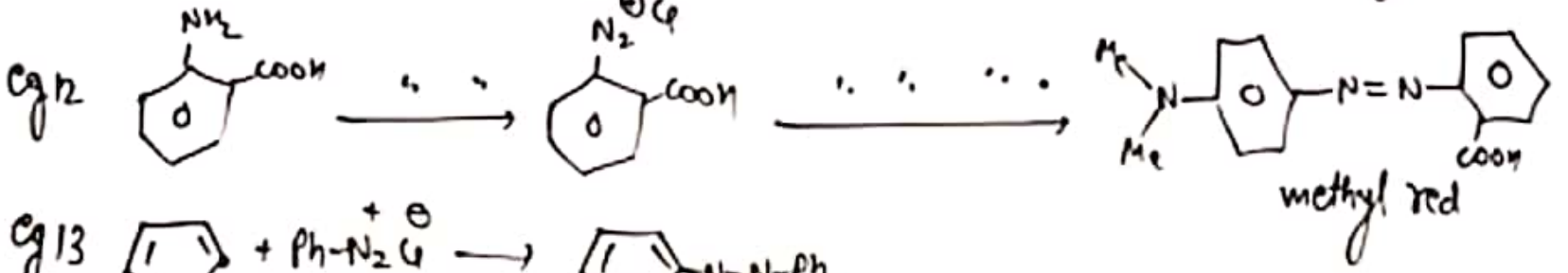
eg 10



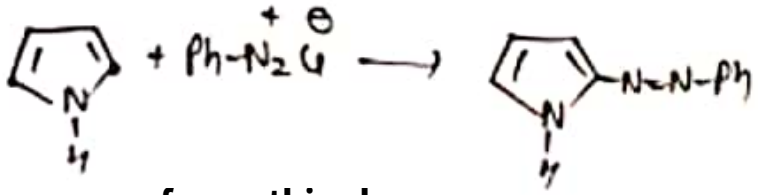
eg 11



eg 12



eg 13



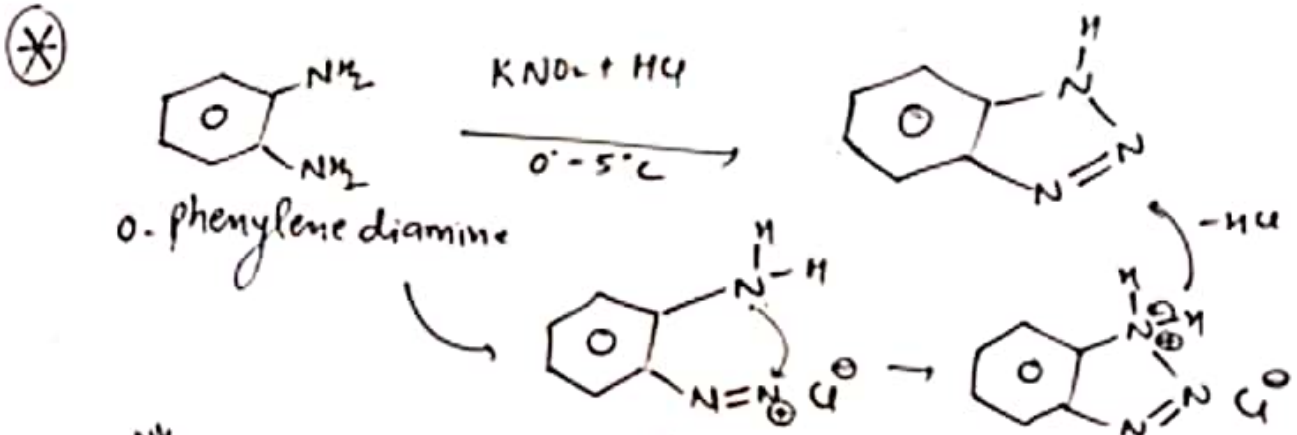
furan, thiophene
do not give coupling rex.
due to lesser reactivity

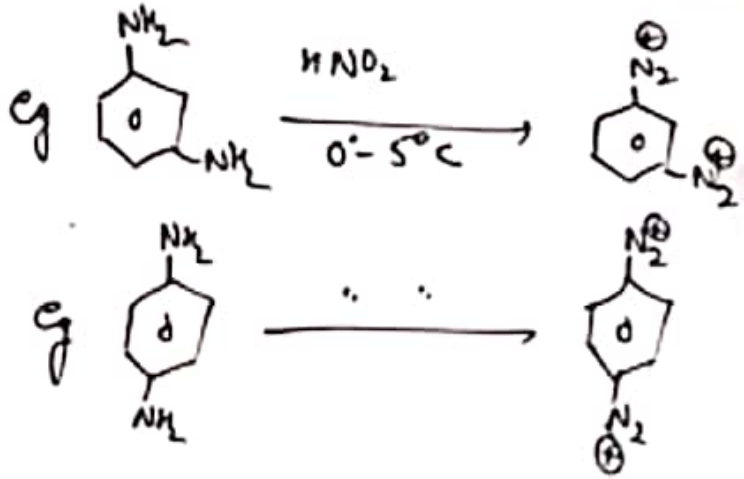
Scanned with CamScanner

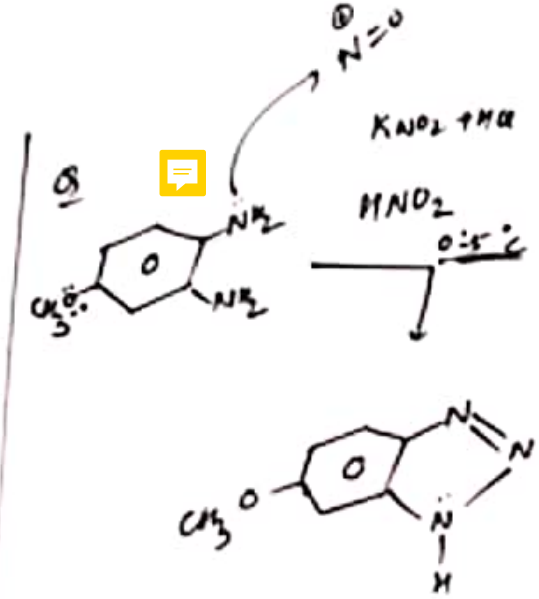
AROMATIC COMPOUNDS

(IV) Benzene diazonium salt:-

(15) Coupling Reaction:-







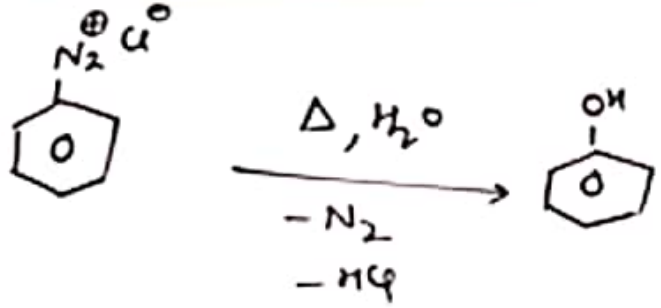
Scanned with CamScanner

AROMATIC COMPOUNDS

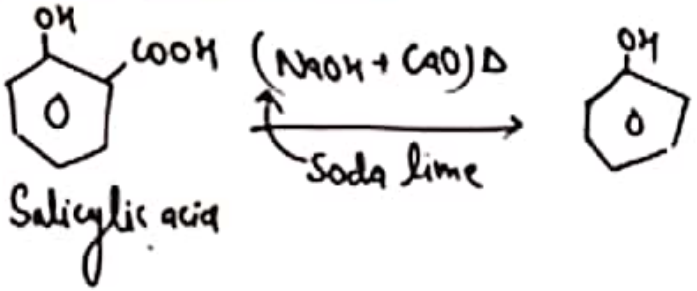
(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

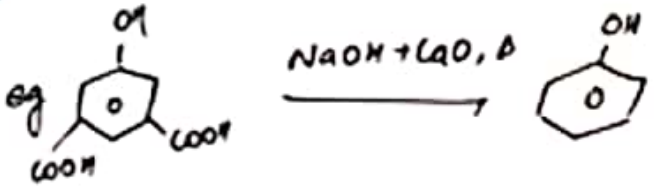
(1) From benzenediazonium salt:-



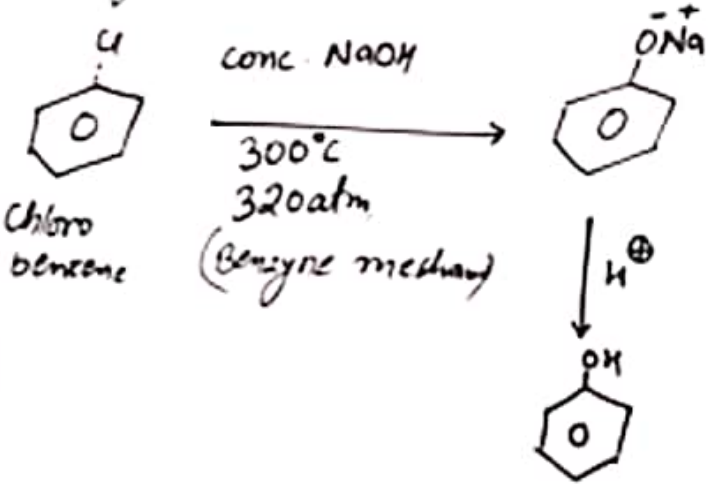
(2) By Decarboxylation of salicylic acid:-



Salicylic acid



(3) By Dow's Process:-



Chloro benzene

Scanned with CamScanner

HKUMHTIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

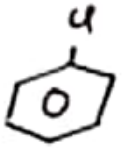
(4) From Grignard Reagent:-

$$RMgX + O_2 \xrightarrow[(air)]{} R-O-O-MgX$$

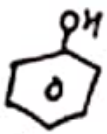
$$R-O-O-MgX + XMg-R \longrightarrow 2\,RO\,MgX$$

$$2\,RO\,MgX \xrightarrow{H_2O+H^{\oplus}} 2\,ROH + 2\,Mg\begin{matrix}OH\\X\end{matrix}$$

eg1 

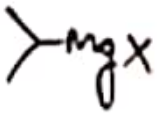
$\xrightarrow{Cl_2 + AlCl_3}$ 

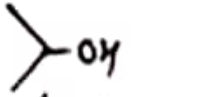
$\xrightarrow{Mg + ether}$  G.R

$\xrightarrow[(ii)\,H_3O^{\oplus}]{(i)\,O_2}$ 

Phenol

eg2 

$\xrightarrow{Mg\ ether}$ 

$\xrightarrow[(ii)\,H_2O]{(i)\,O_2}$ 

Alcohol

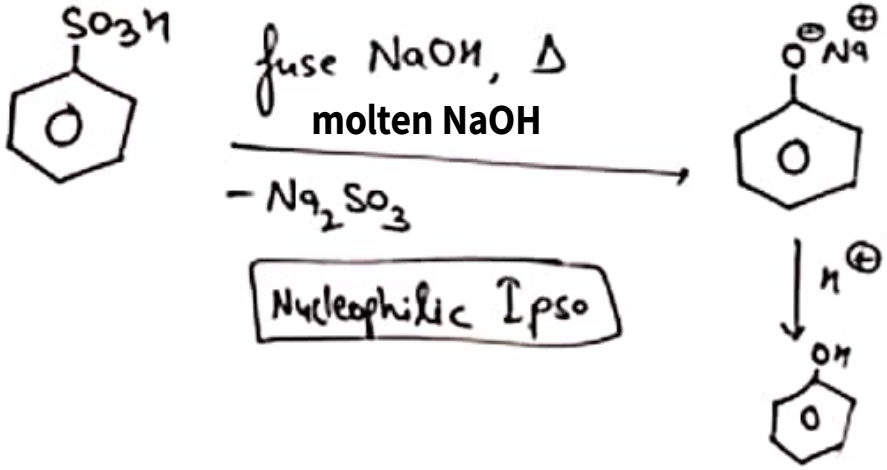
Scanned with CamScanner

AROMATIC COMPOUNDS

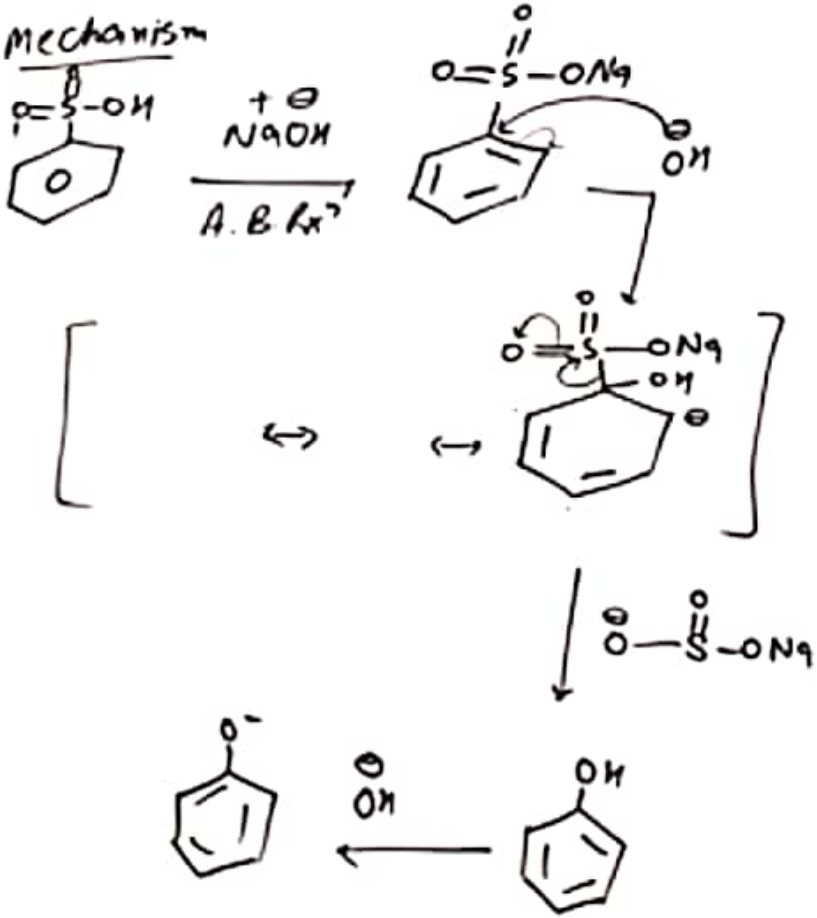
(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

(5) From benzene sulphonic acid :-



Mechanism



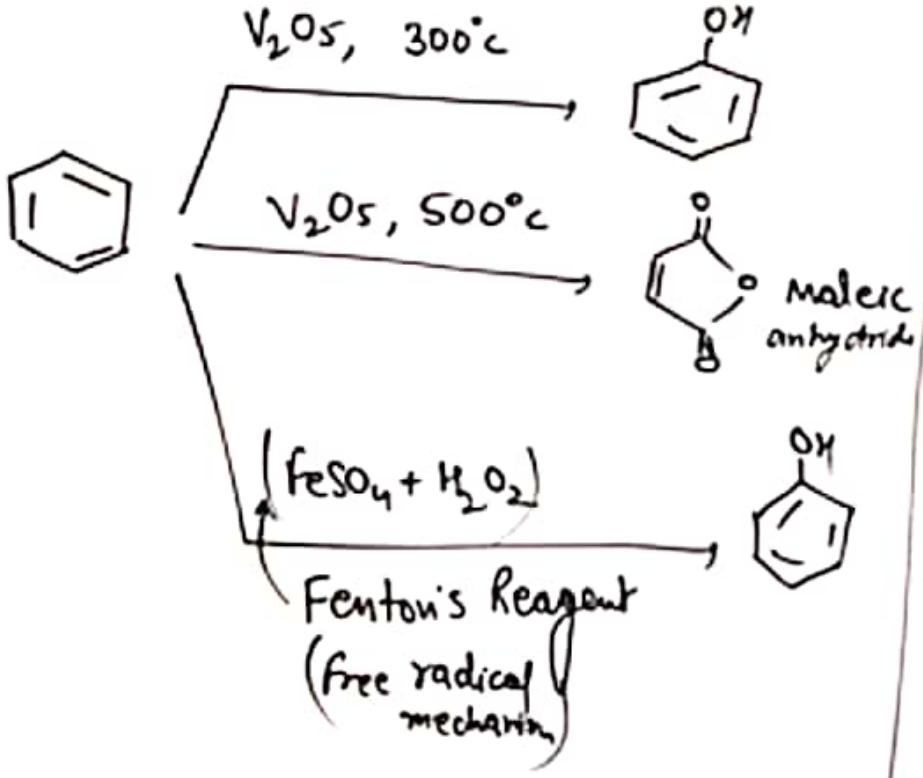
Scanned with CamScanner

AROMATIC COMPOUNDS

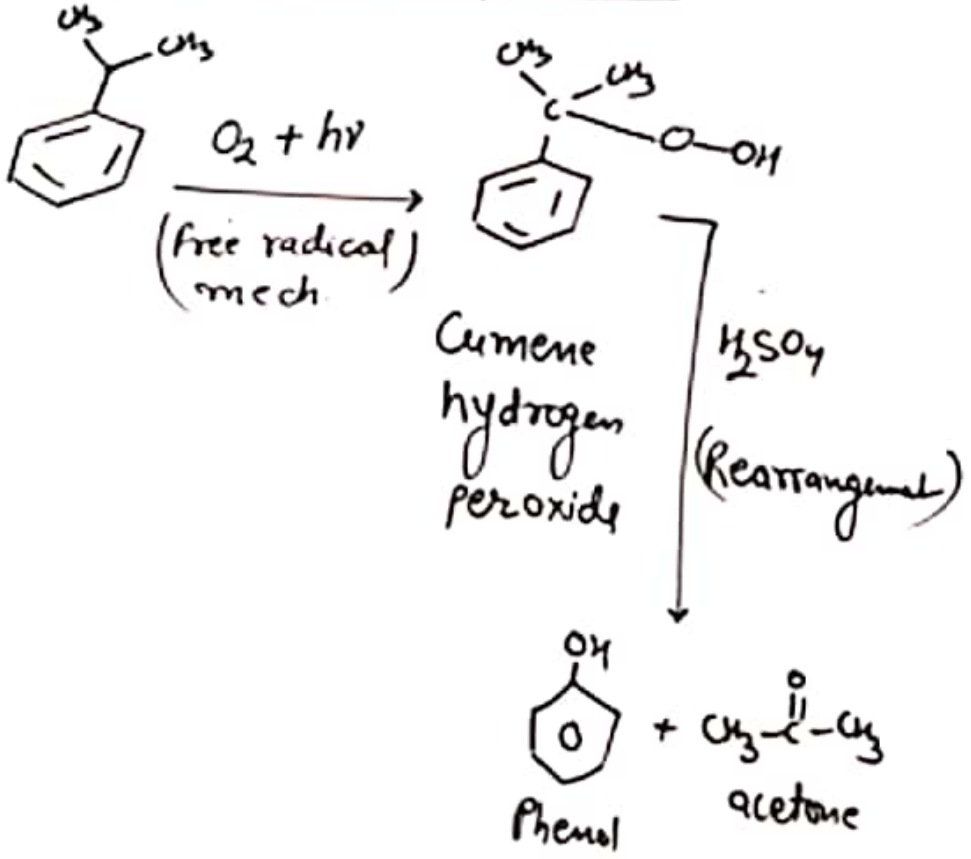
(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

(6) From Benzene



(7) From Cumene = Isopropylbenzene :-



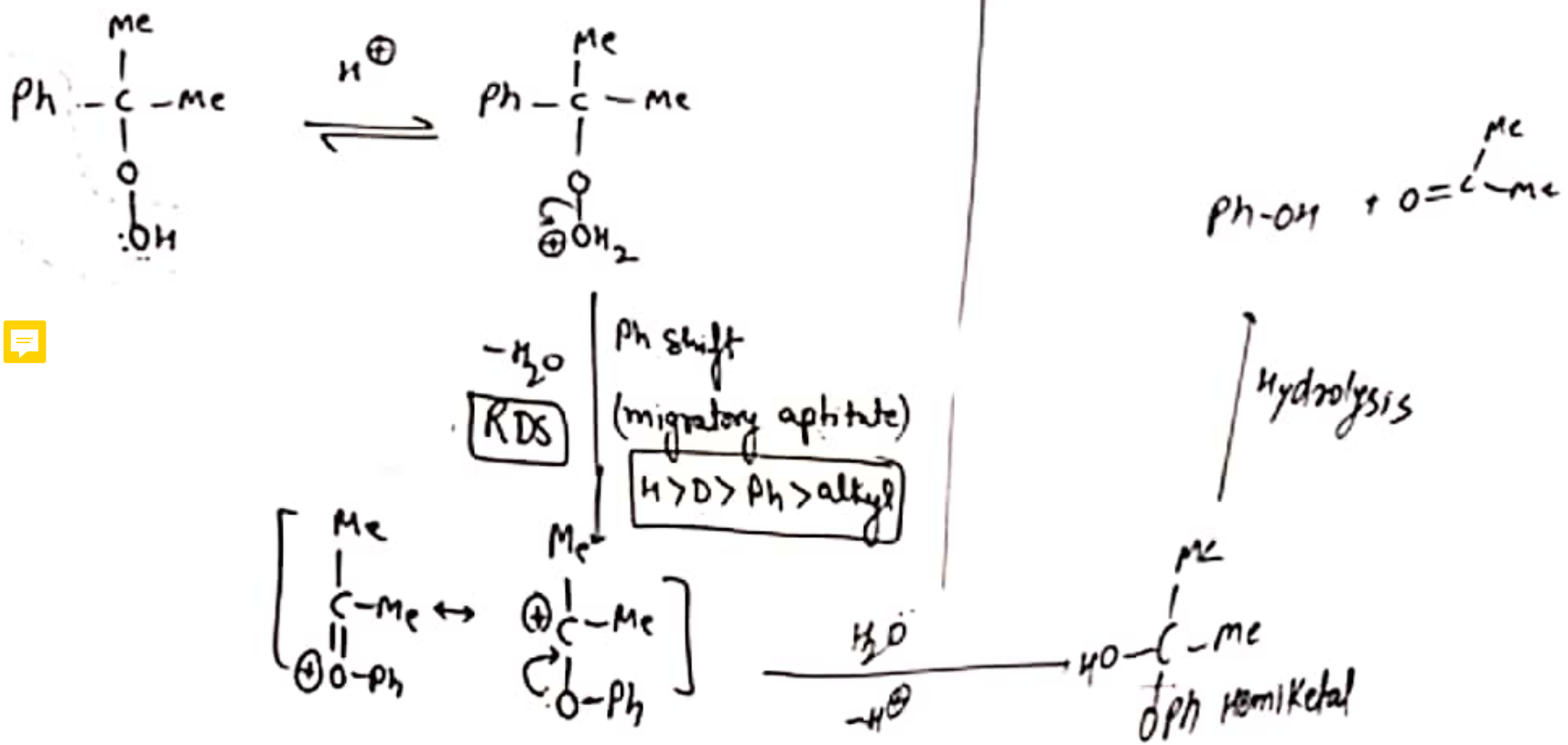
Scanned with CamScanner

AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

Mechanism



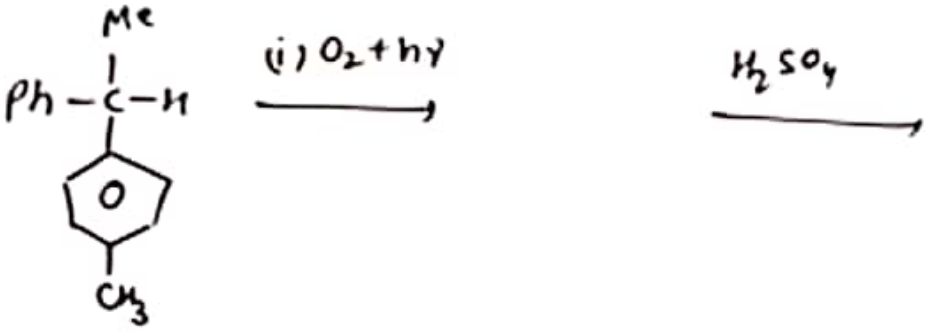
Scanned with CamScanner

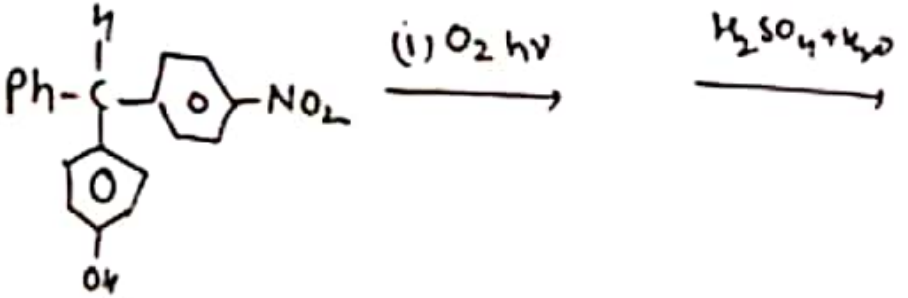
AROMATIC COMPOUNDS

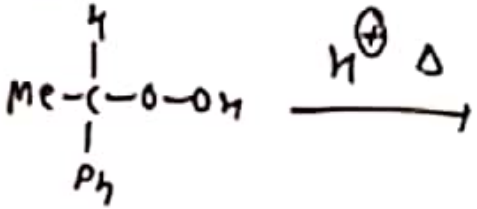
(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

Q

① 

② 

③ 

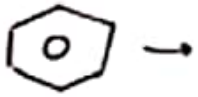
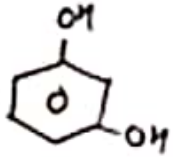
Scanned with CamScanner

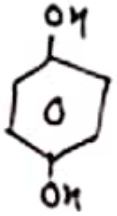
AROMATIC COMPOUNDS

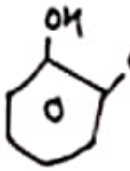
(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

Q Interconversion:-

①  →  Resorcinol

②  →  Hydroquinone

③  →  Catechol

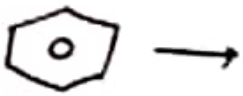
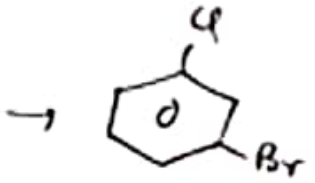
Scanned with CamScanner

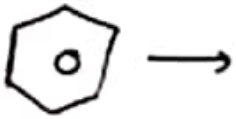
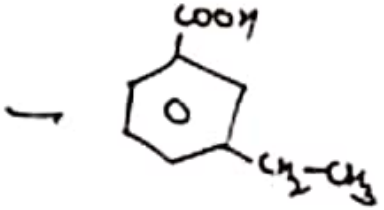
AROMATIC COMPOUNDS

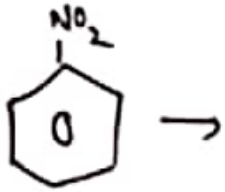
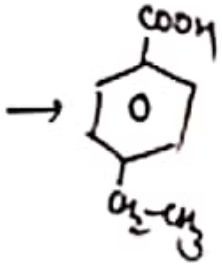
(V) PHENOL (Carbolic acid)

Methods of formation of Phenol:-

Q Interconversion:-

④  → 

⑤  → 

⑥  → 

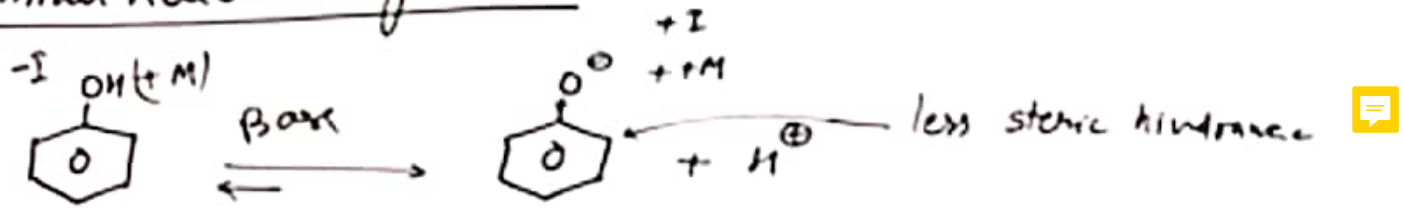
check solutions in screenshots

Scanned with CamScanner

AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.



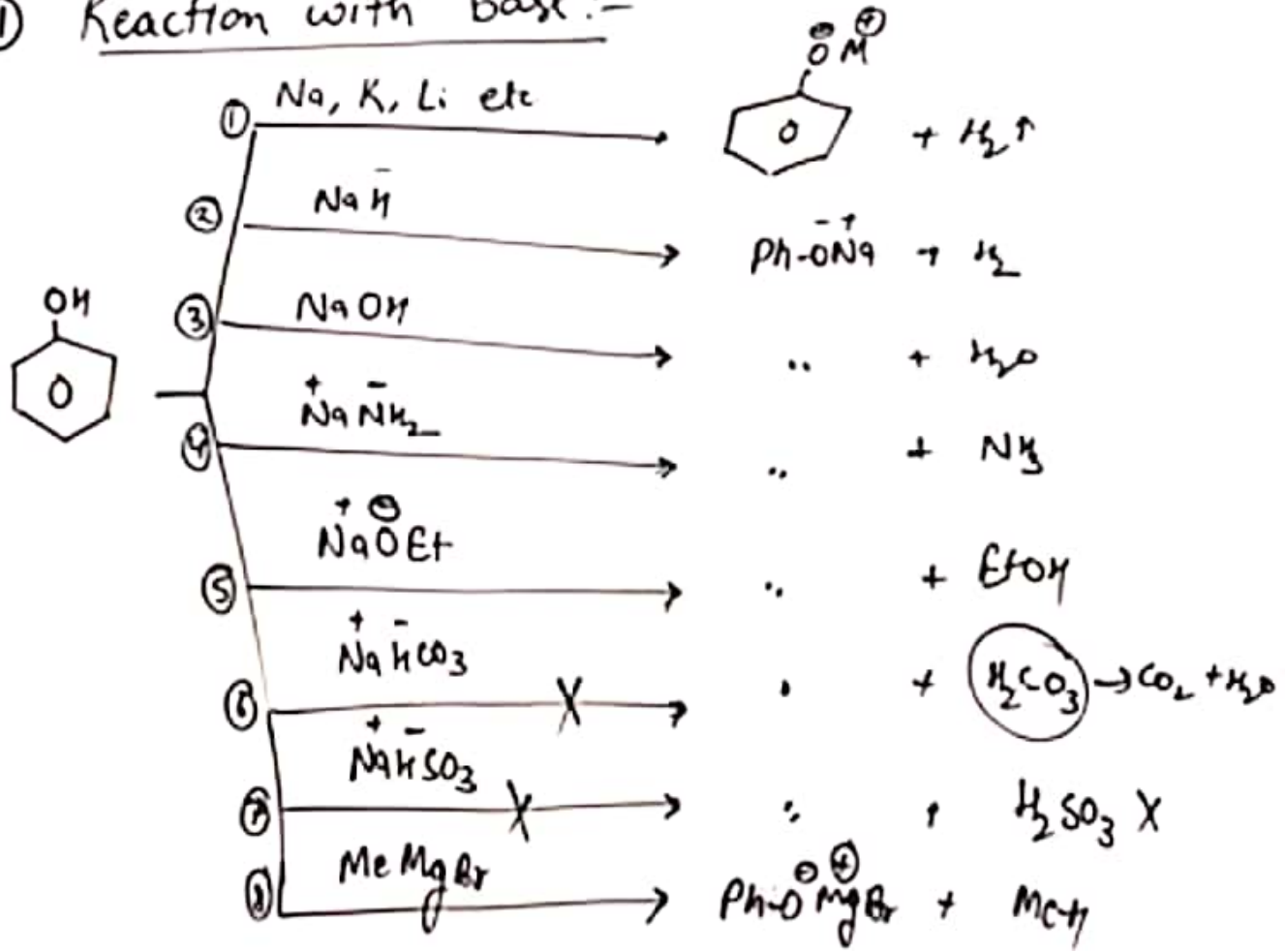
Scanned with CamScanner

AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol

① Reaction with base :-



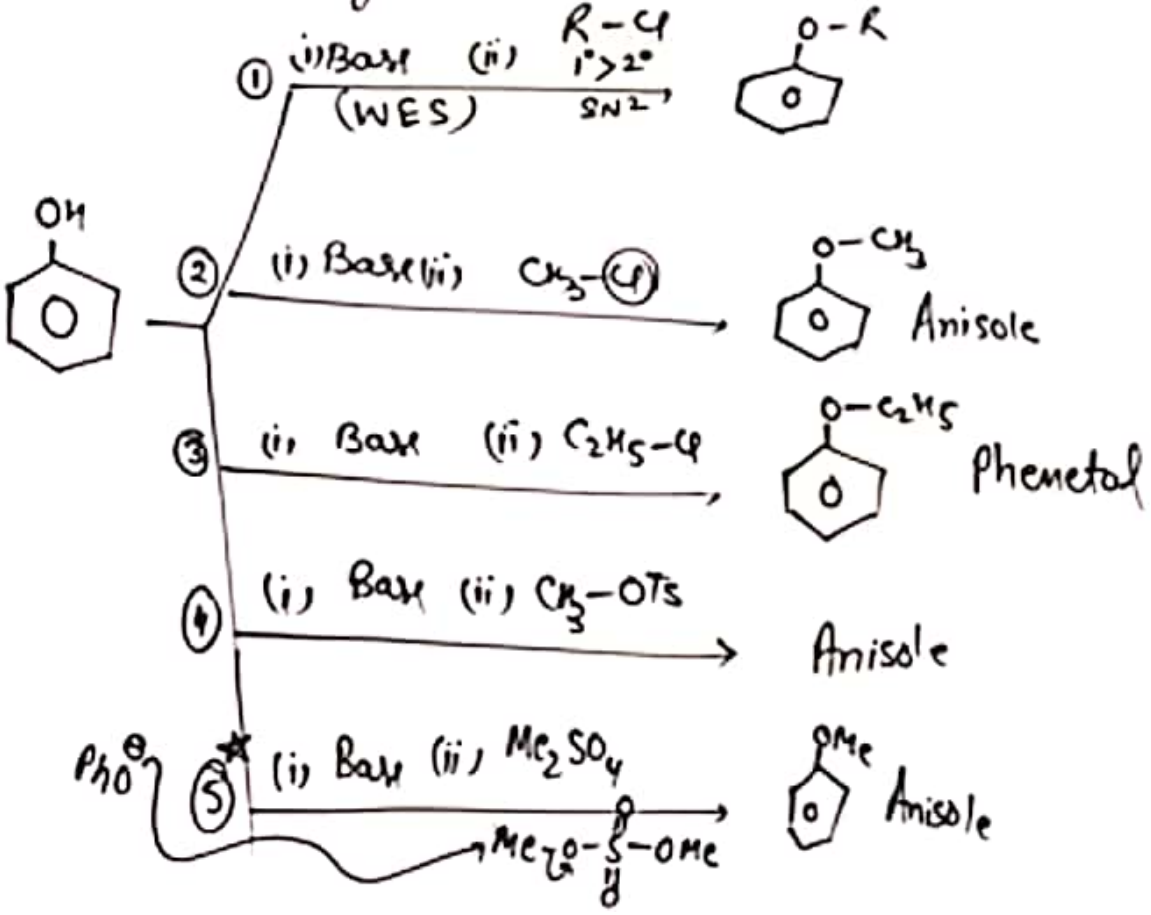
Scanned with CamScanner

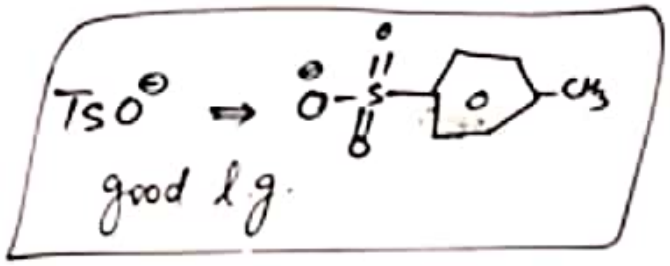
AROMATIC COMPOUNDS

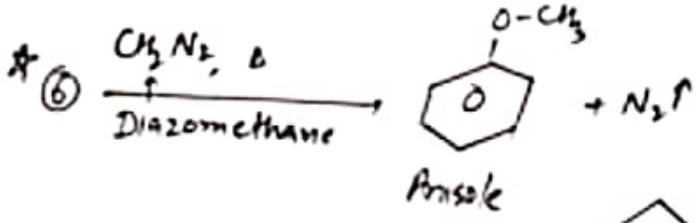
(V) PHENOL (Carbolic acid)

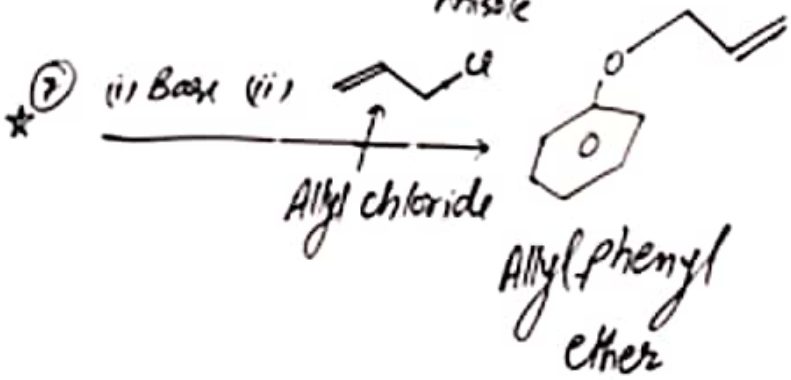
Chemical Reactions of Phenol.

(2) Formation of ethers with Phenol









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AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with phenol

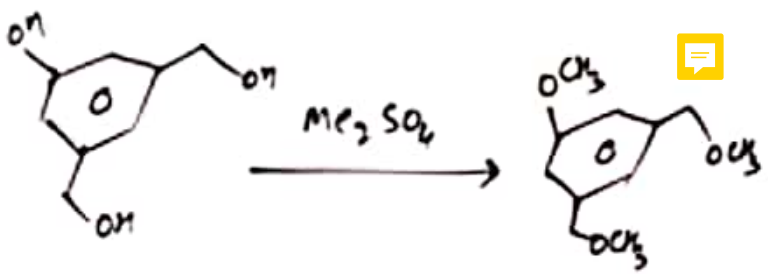
Note 1 Dimethyl sulphate (Me_2SO_4) is good methylating agent for alcoholic, phenolic, amine and thiol groups:-

$$R-\ddot{O}-H \xrightarrow{Me_2SO_4} R-O-CH_3$$

$$Ar-OH \xrightarrow{\quad\cdot\quad} Ar-O-CH_3$$

$$R-\ddot{N}H_2 \xrightarrow{\quad:\quad} R-NH-Me$$

$$R-SH \xrightarrow{\quad:\quad} R-S-Me$$

eg 

eg 

Scanned with CamScanner

AROMATIC COMPOUNDS

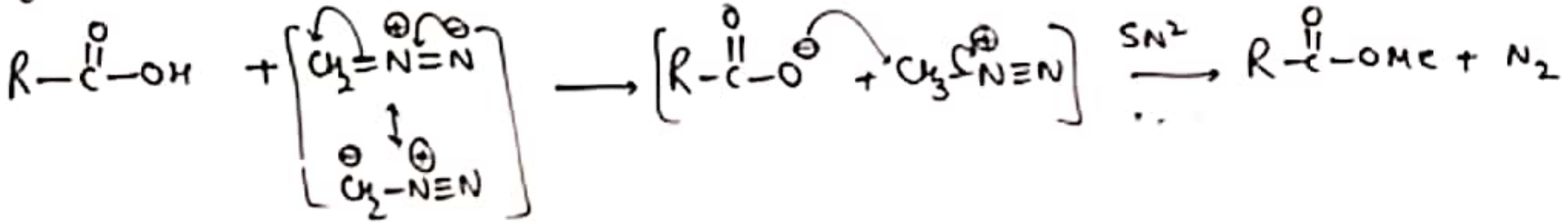
(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with phenol

Note 2 Diazo methane CH_2N_2 is also a good methylating agent for acidic hydrogen containing group. ($-SO_3H$, $-COOH$, $Ar-OH$, $R-OH$ etc)

for less hindered alcohols



R.S. of Diazomethane

Me-N=N-Me
Azomethane

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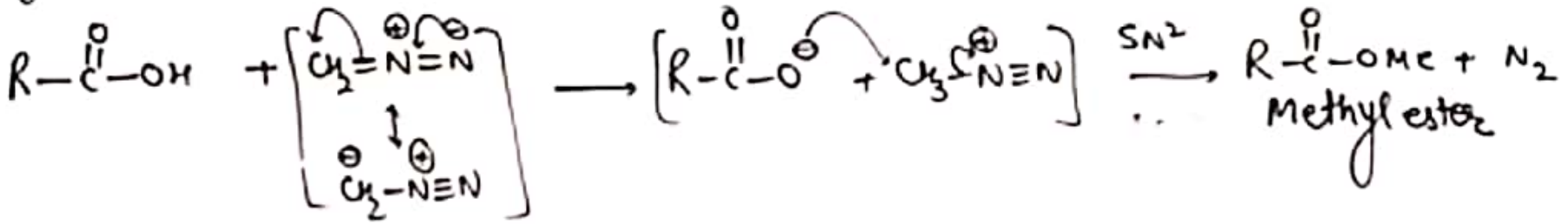
AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with phenol

Note 2 Diazomethane CH_2N_2 is also a good methylating agent for acidic hydrogen containing group. ($-SO_3H$, $-COOH$, $Ar-OH$, $R-OH$ etc)



R.S. of Diazomethane

Me-N=N-Me
Azomethane

Scanned with CamScanner

AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with Phenol

Note 2 Diazo methane CH_2N_2 is also a good methylating agent for acidic hydrogen containing group. ($-SO_3H$, $-COOH$, $Ar-OH$, $R-OH$ etc)

$$R-SO_3H \xrightarrow{CH_2N_2} R-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-OCH_3$$

$$R-COOH \longrightarrow R-\overset{\overset{O}{\|}}{C}-OCH_3$$

$$Ar-OH \longrightarrow Ar-O-CH_3$$

$$R-OH \longrightarrow R-O-CH_3$$

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AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol

(2) Formation of ethers with Phenol

Note 2 Diazomethane CH_2N_2 is also a good methylating agent for acidic hydrogen containing group. ($-SO_3H$, $-COOH$, $Ar-OH$, $R-OH$ etc)

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$$R-COOH \longrightarrow R-\overset{\overset{O}{\|}}{C}-OCH_3$$

$$Ar-OH \longrightarrow Ar-O-CH_3$$

$$R-OH \longrightarrow R-O-CH_3$$

Scanned with CamScanner

HROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

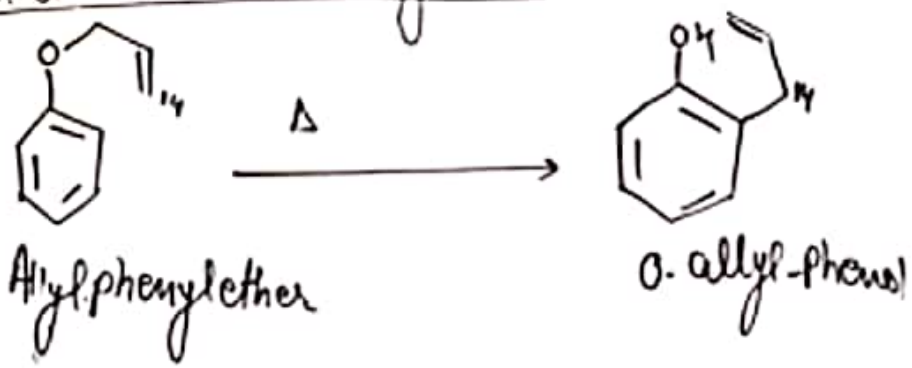
Chemical Reactions of Phenol.

$CH_2=CH-CH_2-$ allyl group

(2) Formation of ethers with phenol

Note 3. Allyl-aryl ether give Claisen Rearrangement on high temp and form ortho-allyl phenol. This rearrangement is also given by allyl-vinyl ethers

⇒ "Claisen Rearrangement": (Intramolecular rearr-)



Allylphenylether → O-allyl-Phenol

Scanned with CamScanner

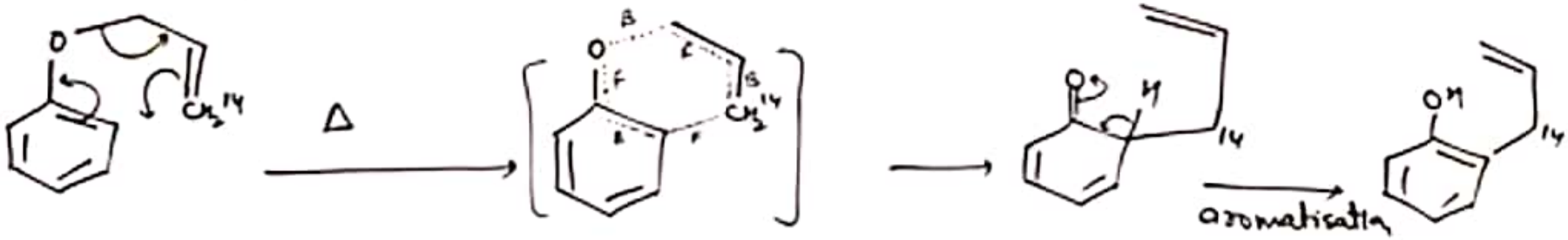
HROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with phenol

Mechanism of chaise Rearrang.. a Pericyclic mech.



Scanned with CamScanner

HROMATIC COMPOUNDS

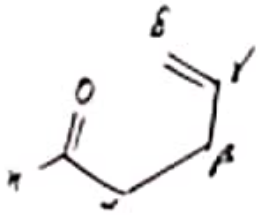
(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with phenol

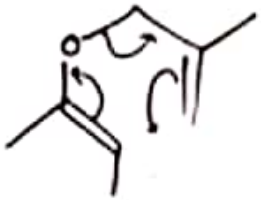
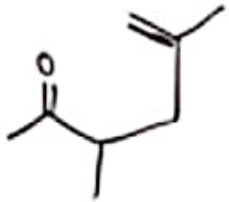
Mechanism of chaise Rearrang. = Pericyclic mech.

eg1 

$\xrightarrow[\text{(Claisen rearr.)}]{\Delta}$ 6 MCTS $\longrightarrow$ 

γ, δ-Unsaturated carbonyl comp

Allyl-vinyl ether

eg 2  $\xrightarrow{\Delta}$ 

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AROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

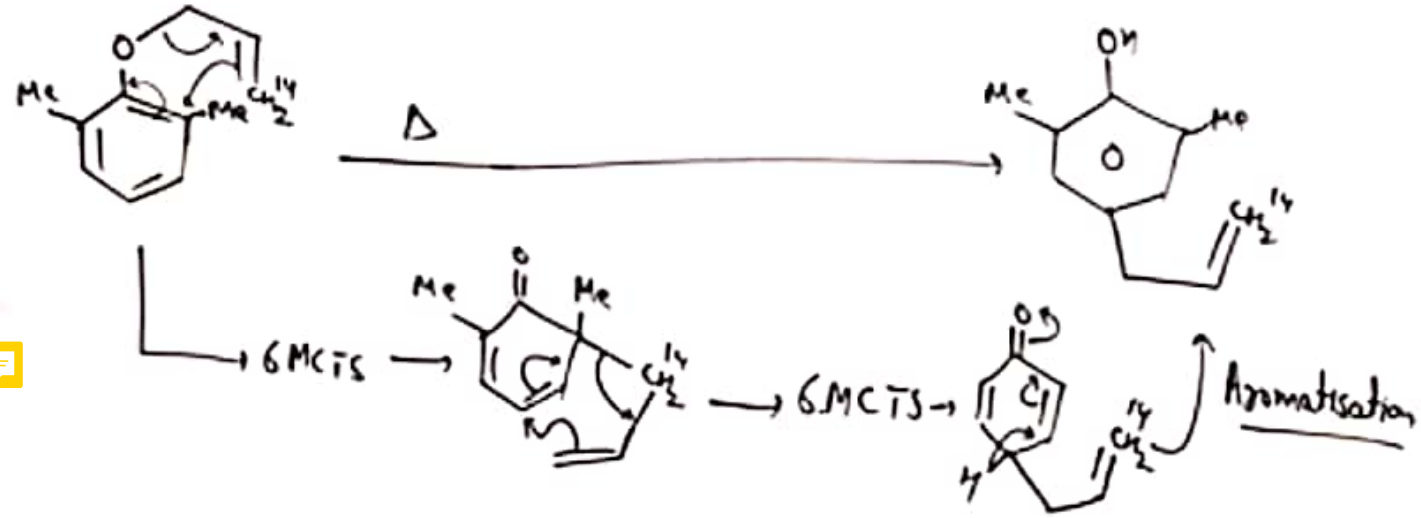
(2) Formation of ethers with phenol

Mechanism of Chaise Rearrang. → Pericyclic mech.

eg 3



* eg 4



Scanned with CamScanner

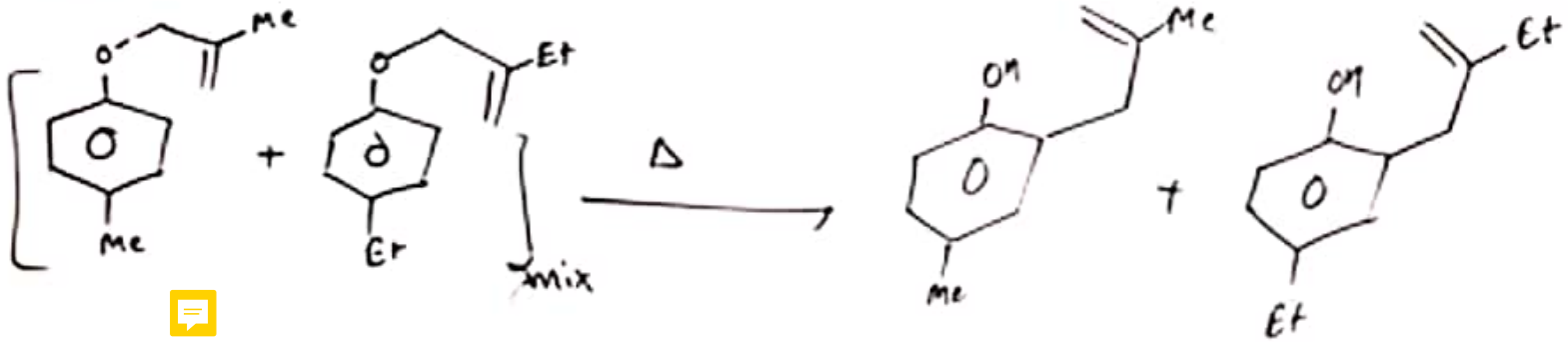
HROMATIC COMPOUNDS

(V) PHENOL (Carbolic acid)

Chemical Reactions of Phenol.

(2) Formation of ethers with Phenol

Note: Claisen Rearrangement



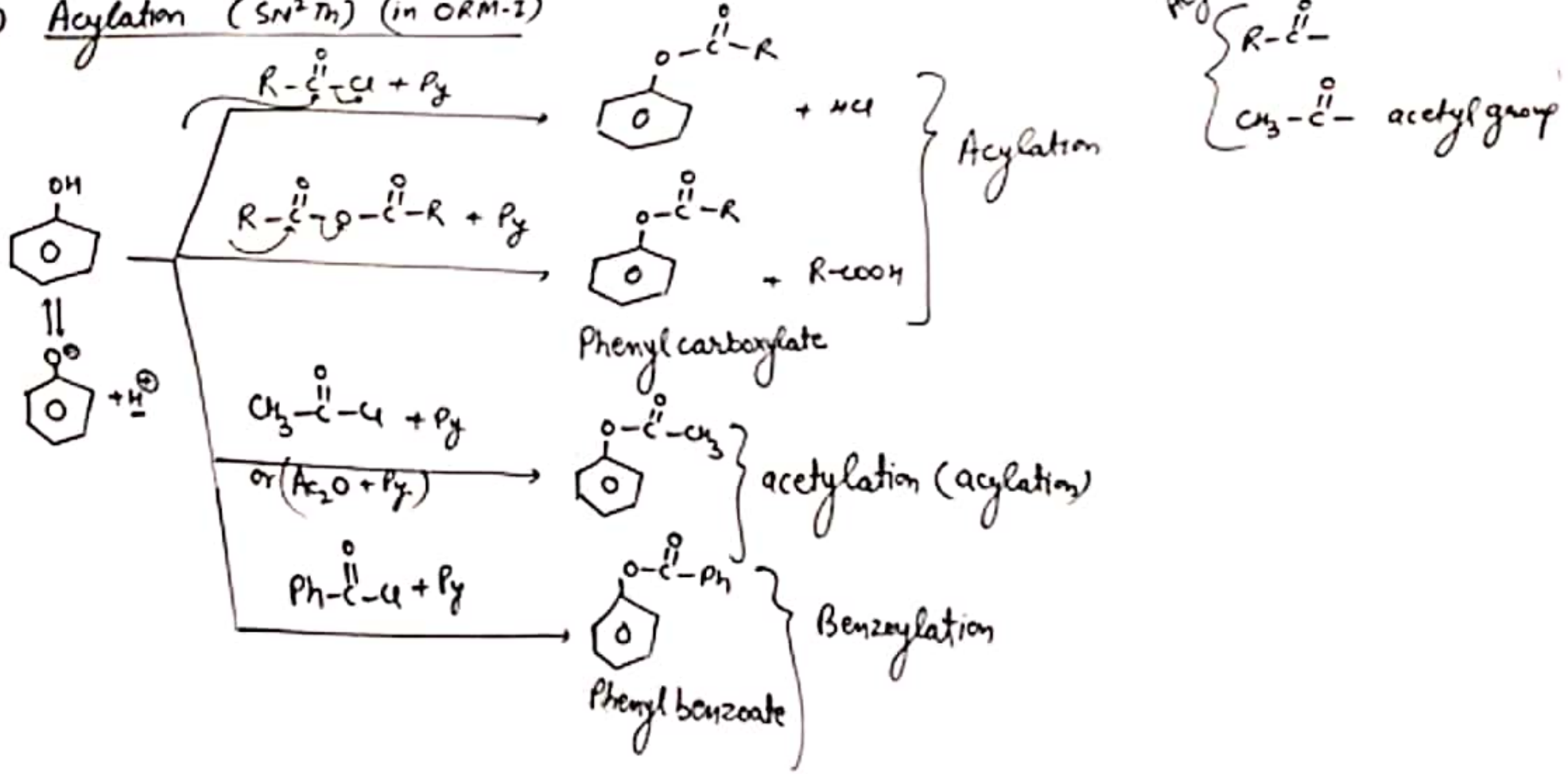
formation of only 2 products is evidence of intramolecular rex.

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AROMATIC COMPOUNDS

(V) PHENOL (carbolic acid) ⇒ Chemical Reactions of Phenol

(3) Acylation (SN^2 m) (in ORM-2)



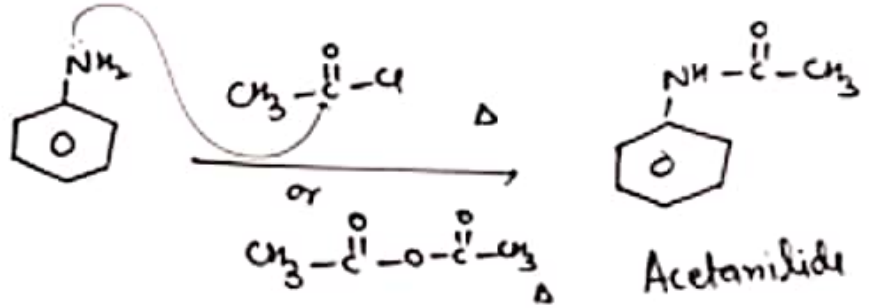
Scanned with CamScanner

AROMATIC COMPOUNDS

(V) PHENOL (carbolic acid) ⇒ Chemical Reactions of Phenol

(3) Acylation (SN^2 rn) (in ORM-1)

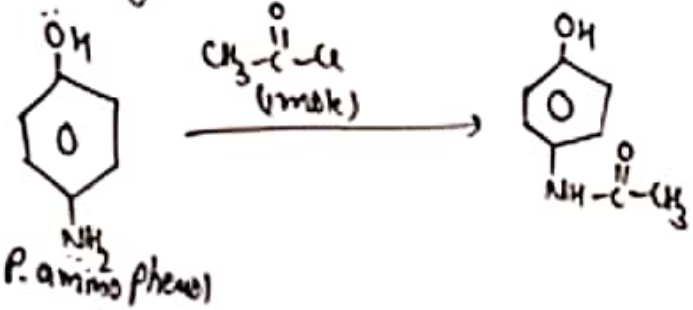
Note: 1. This reaction is also given by aniline (amine)



Acetanilide
(N-Phenyl acetamide)

Note 2 Benzoylation of phenol and aniline is known as "Schotten-Baumann's Reaction"

eg



P. amino phenol

Paracetamol (Antipyretic ⇒ Reduce fever)
(Analgesic ⇒ Pain Killer)

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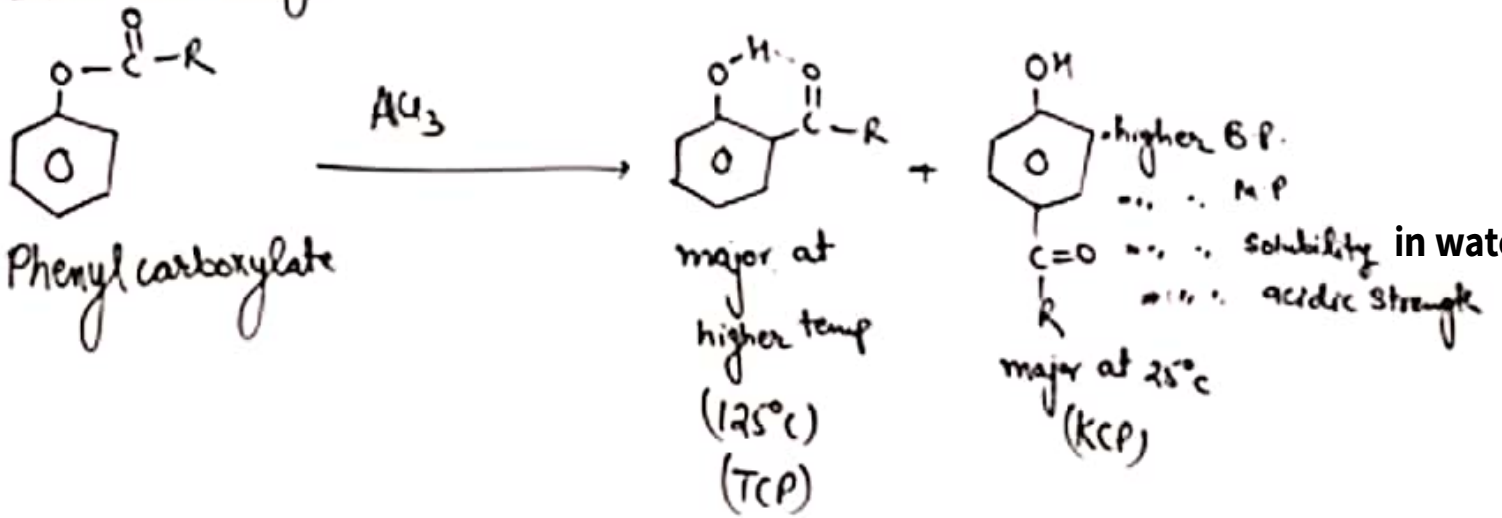
AROMATIC COMPOUNDS

(V) PHENOL (carbolic acid) ⇒ Chemical Reactions of Phenol

(3) Acylation (SN^2 rn) (in ORM-I)

Note 3 Aryl carboxylate ($R-\overset{O}{\overset{\|}{C}}-O-Ar$) give "Fries Rearrangement" in presence of $AlCl_3$. This rearrangement is intramolecular as well as intermolecular rearr.

⇒ **Fries Rearrangent:-**



Phenyl carboxylate

major at higher temp (125°C) (TCP)

major at 25°C (KCP)

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AROMATIC COMPOUNDS

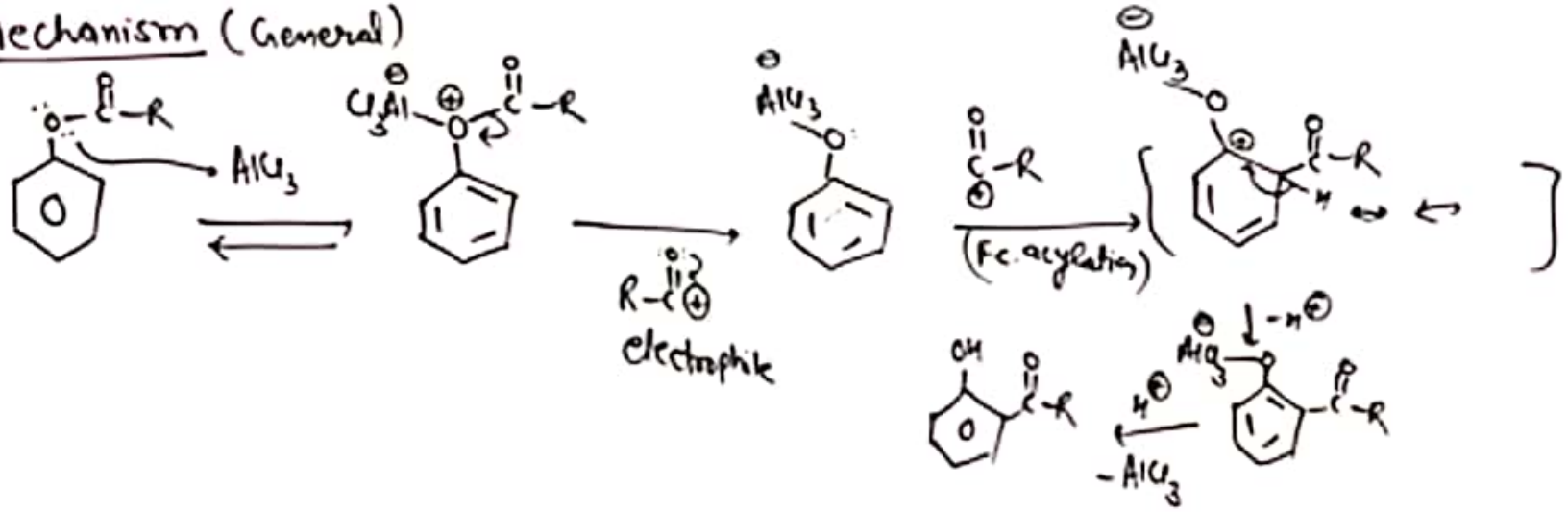
(V) PHENOL (carbolic acid) ⇒ Chemical Reactions of Phenol

(3) Acylation (SN^2 Th) (in ORM-I)

Note 3 Aryl carboxylate ($R-\overset{O}{\overset{\|}{C}}-O-Ar$) give "Fries Rearrangement" in presence of $AlCl_3$. This rearrangement is intramolecular as well as intermolecular rearr.

⇒ **Fries Rearrangent**:-

Mechanism (General)



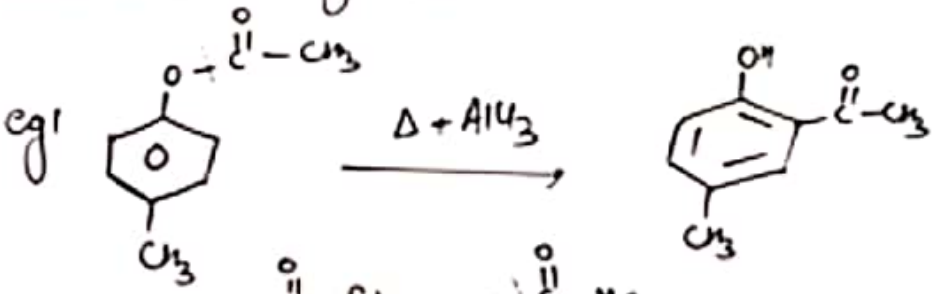
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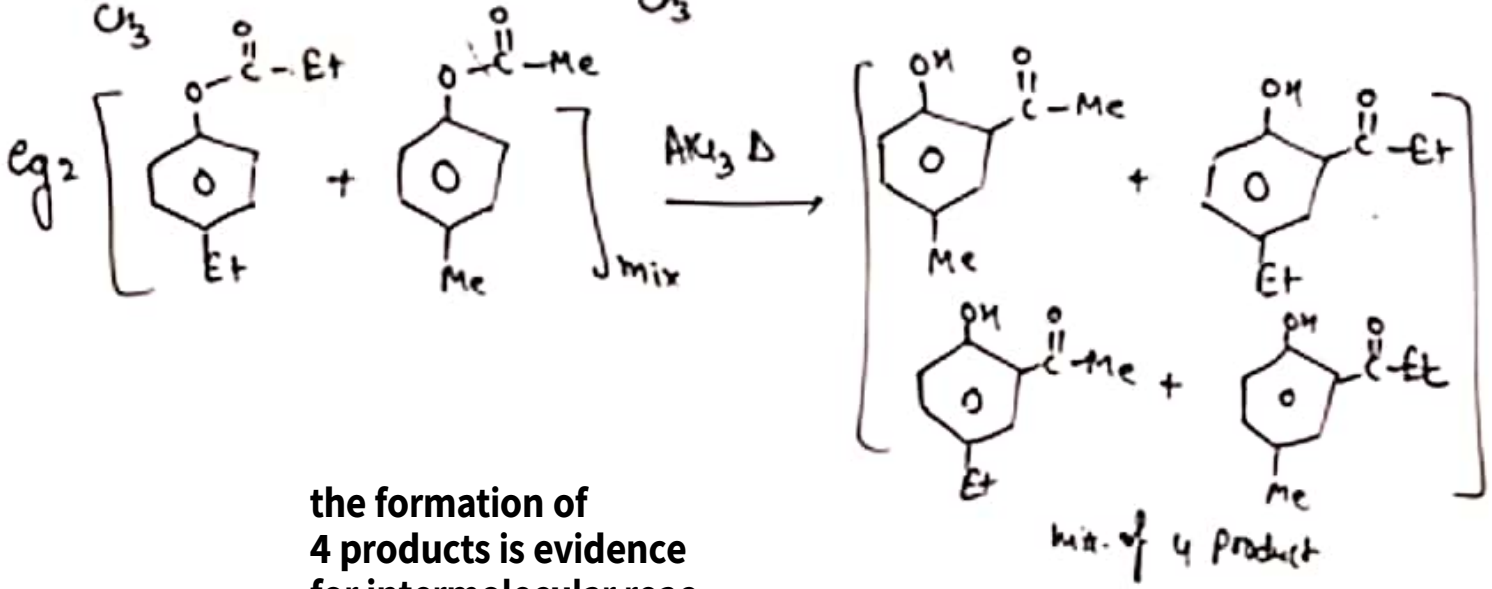
AROMATIC COMPOUNDS

(V) PHENOL (carbolic acid) ⇒ Chemical Reactions of Phenol

(3) Acylation (SN^2 Th) (in ORM-I)

Fries Rearrangement





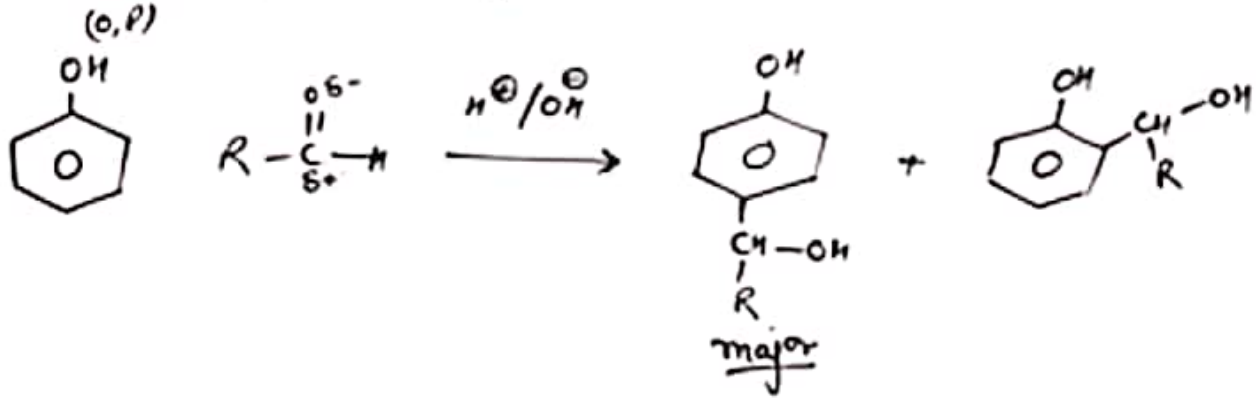
the formation of 4 products is evidence for intermolecular reac.

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AROMATIC COMPOUNDS

(I) PHENOL (carbolic acid)

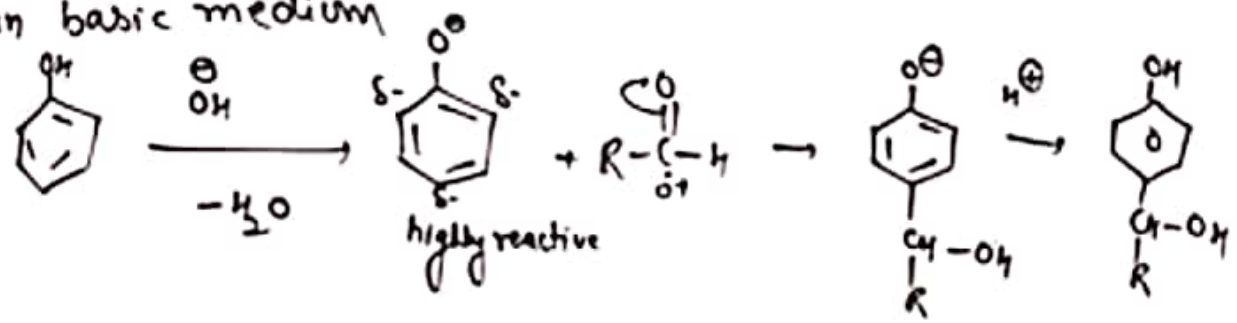
(4) Reaction with aldehydes:-



reaction with ketones is difficult due to steric reasons and +I effect of alkyl group

mechanism

ⓐ in basic medium



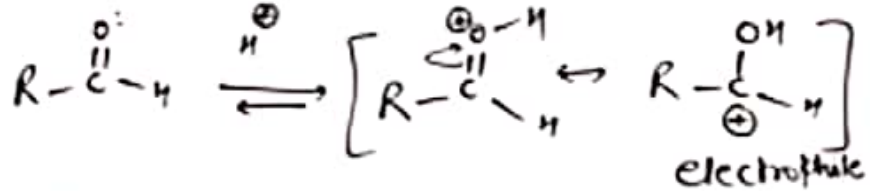
Scanned with CamScanner

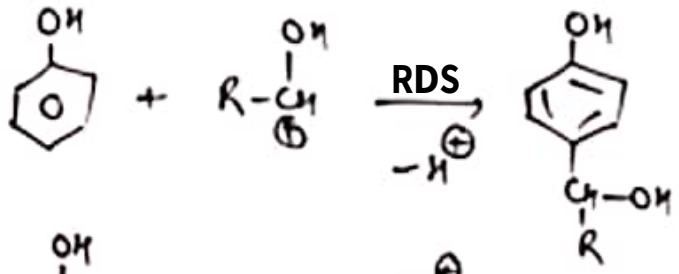
AROMATIC COMPOUNDS

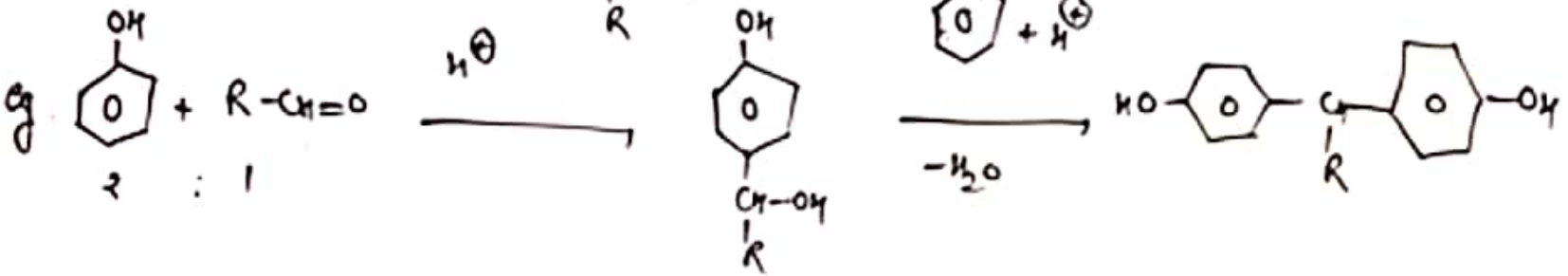
(I) PHENOL (Carbolic acid)

(4) Reaction with aldehydes:-

(b) Mechanism in acid:-







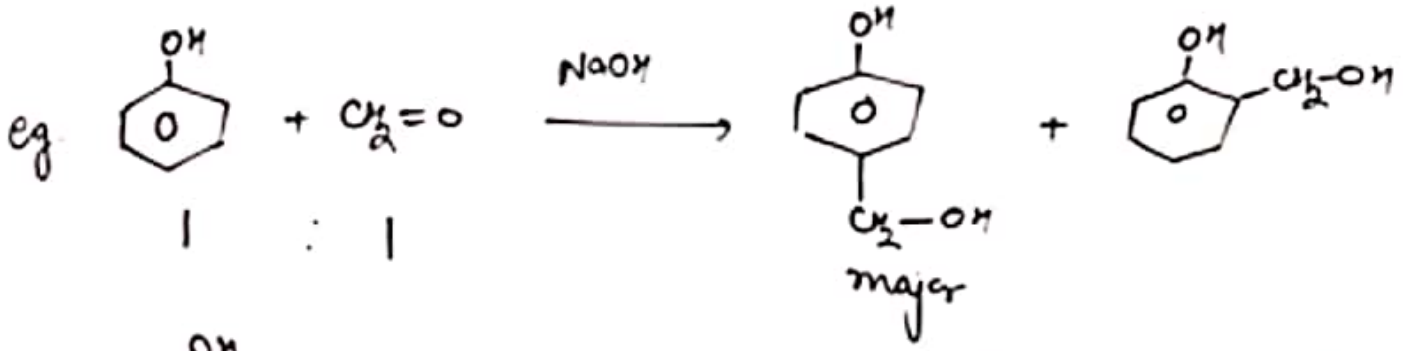
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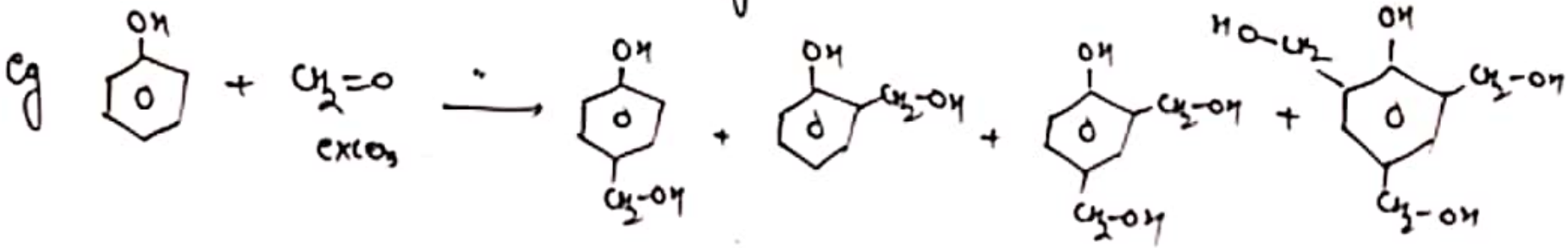
AROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

(4) Reaction with aldehydes:-

→ Lederer - Manasse Reaction:- **reaction b/w phenol and formaldehyde**





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AROMATIC COMPOUNDS

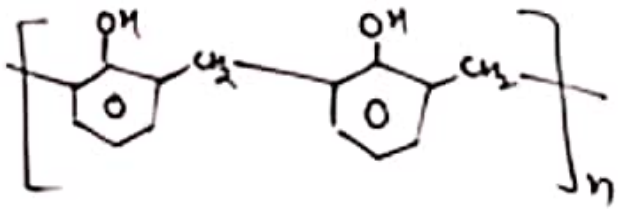
(I) PHENOL (Carbolic acid)

(4) Reaction with aldehydes:-

→ Lederer - Manasse Reaction:-

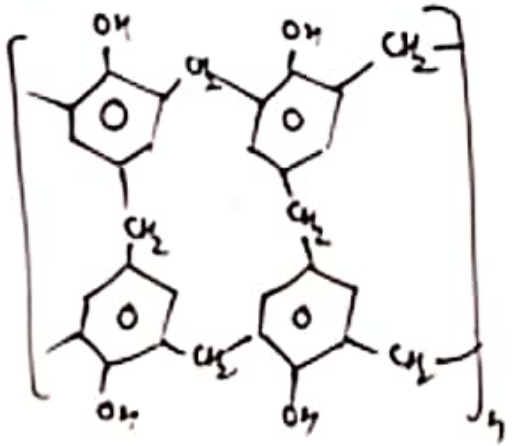
Products of this reaction can form polymers

* Novalac = Resol → Linear polymer:-



(in paint . Adhesive)

⊛ Bakelite = 3D Polymer (cross link polymer)

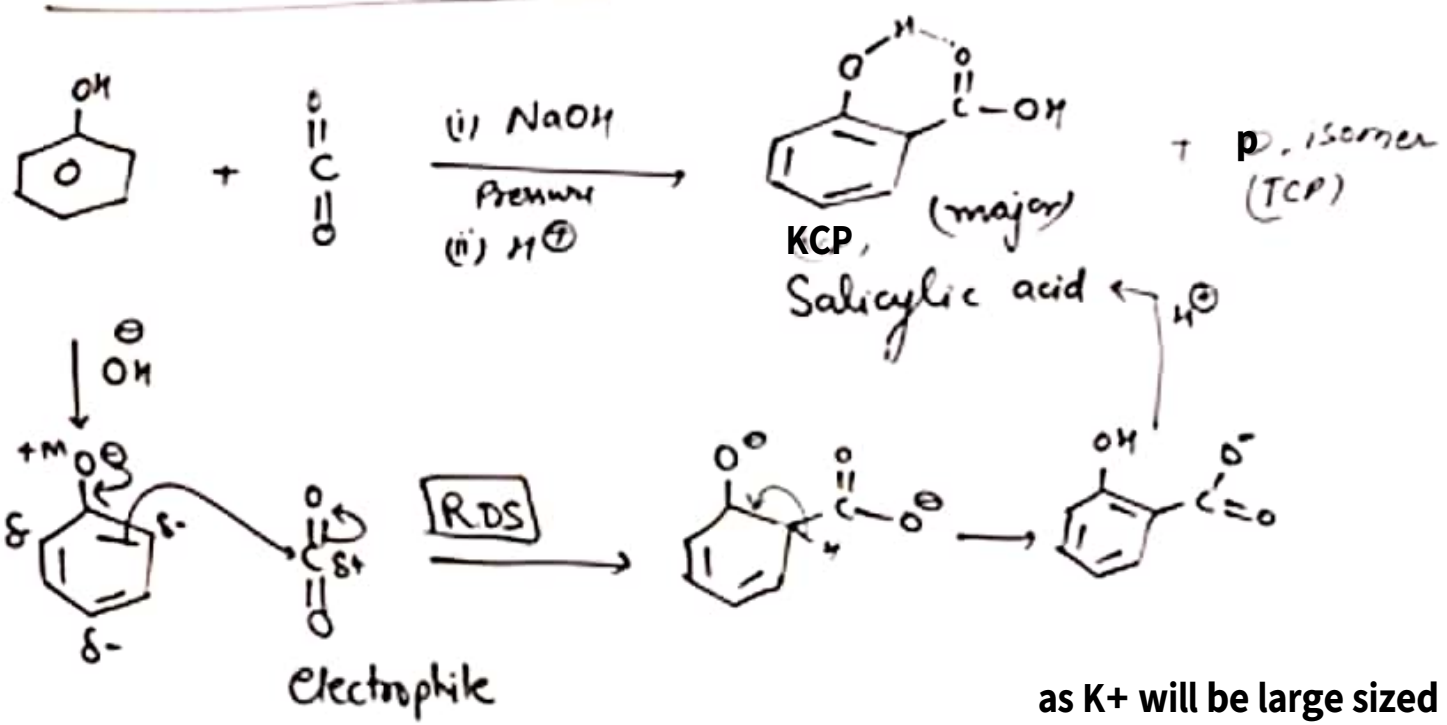


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HROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

(5) Kolbe's - Schmitt Reaction:-



⇒ Note:- in Presence of KOH P. Product - major

as K+ will be large sized and hence will repel the carboxlic acid grp

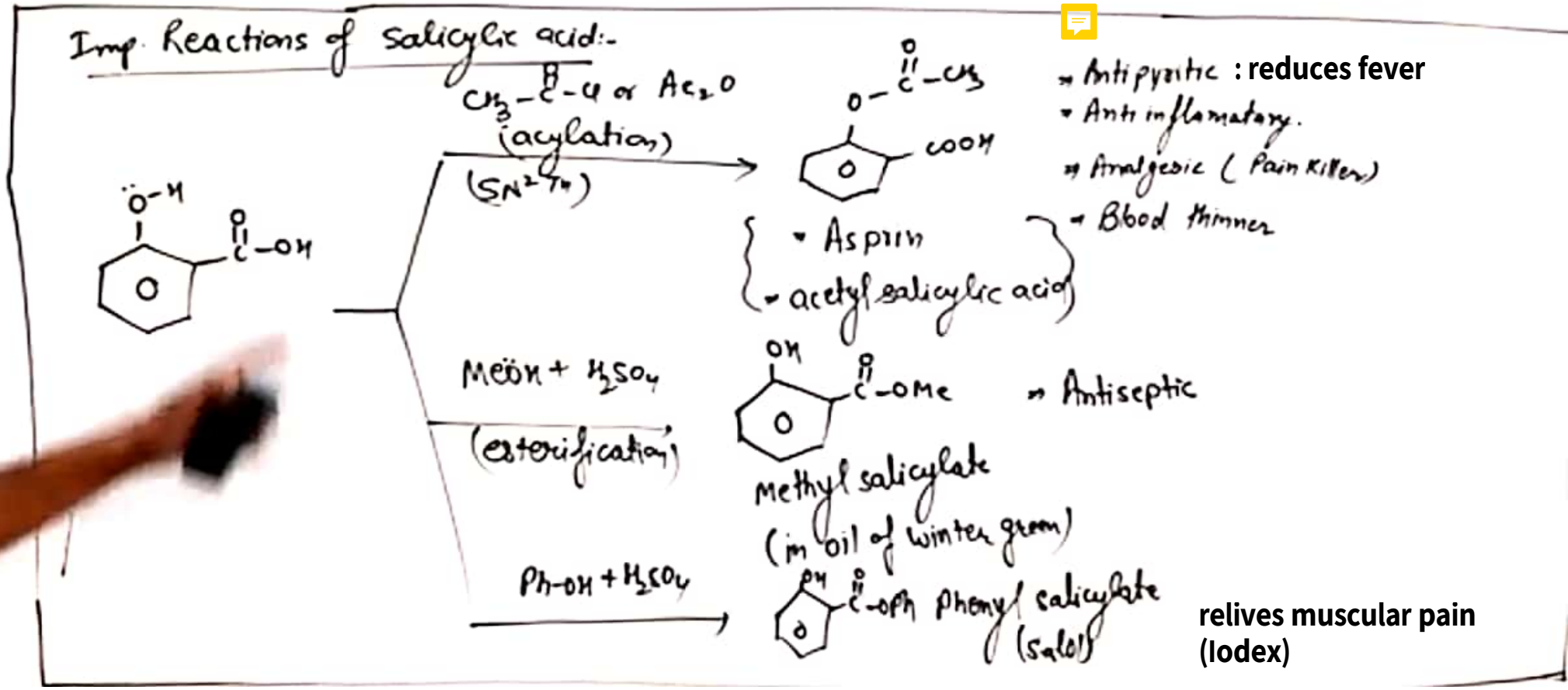
- Salicylic acid is useful starting material for many useful comp.

Scanned with CamScanner

AROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

(5) Kolbe's - Schmitt Reaction:-



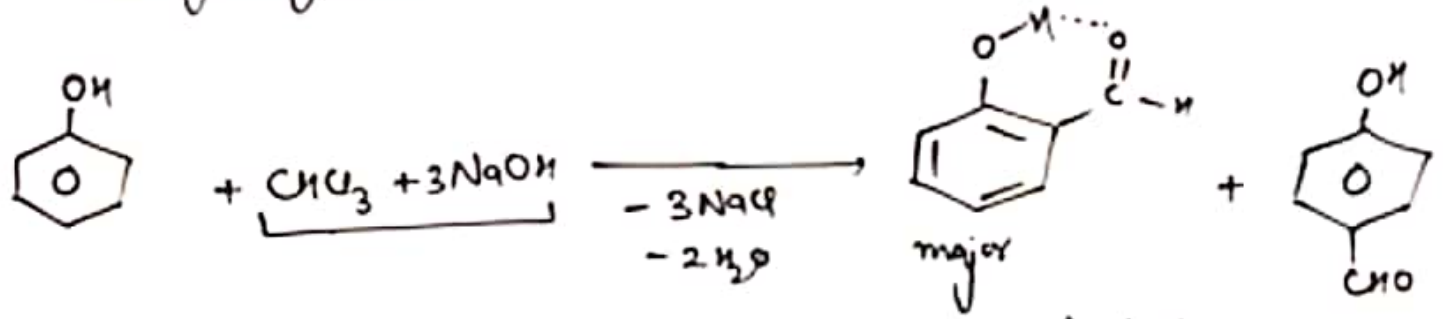
Scanned with CamScanner

HROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

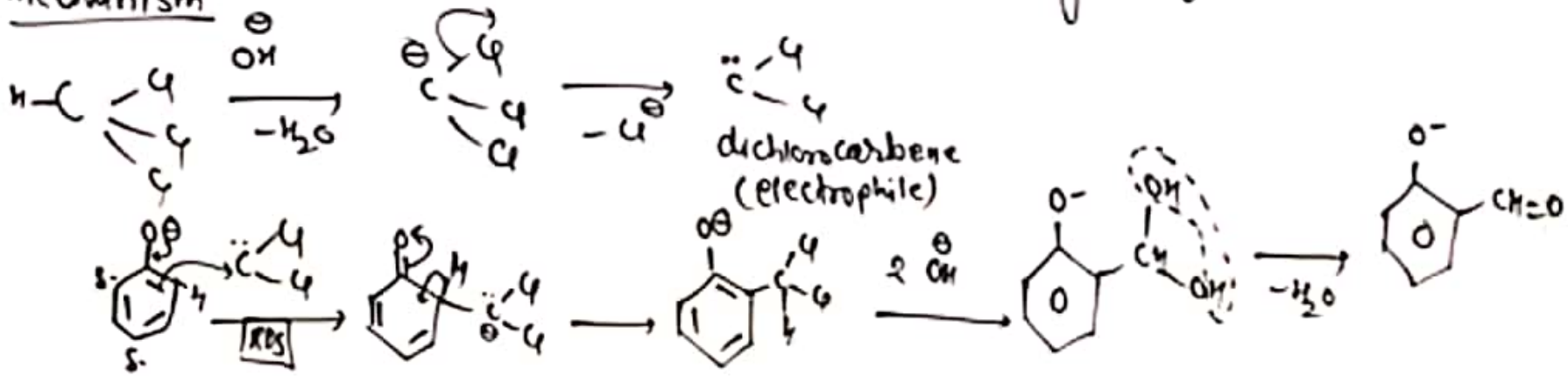
(6) Reimer - Tiemann's Reaction:-

ⓐ R.T. formylation:-



Salicyaldehyde

mechanism



H gets attached
at the most basic position

Scanned with CamScanner

HROMATIC COMPOUNDS

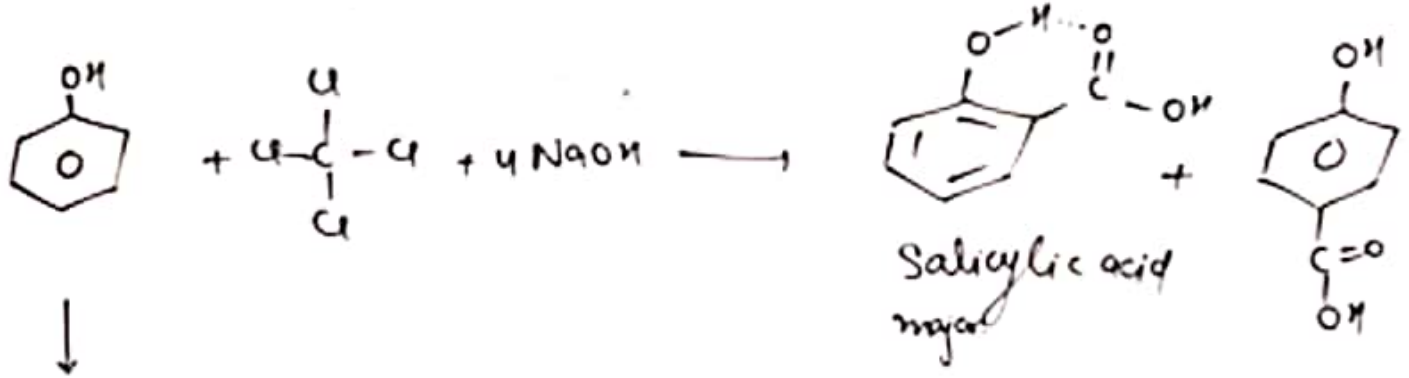
(I) PHENOL (Carbolic acid)

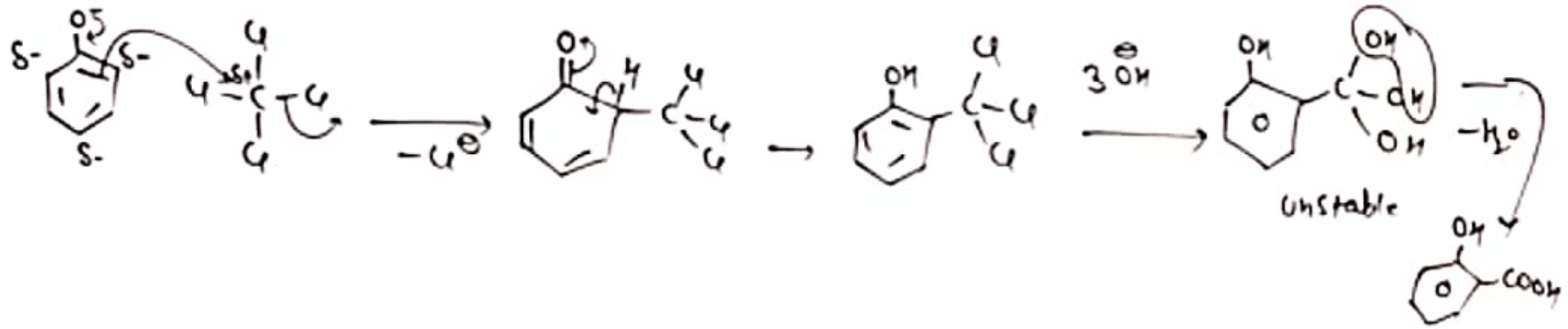
(6) Reimer - Tiemann's Reaction:-

(b) R.T. carboxylation:-

more acidic due to H-bonding

separated by steam distillation as they have a good difference in BP due to H-bonding



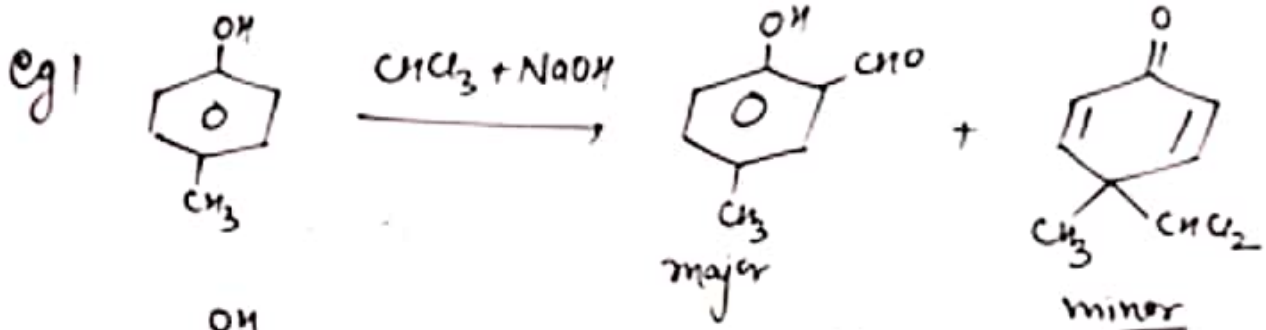


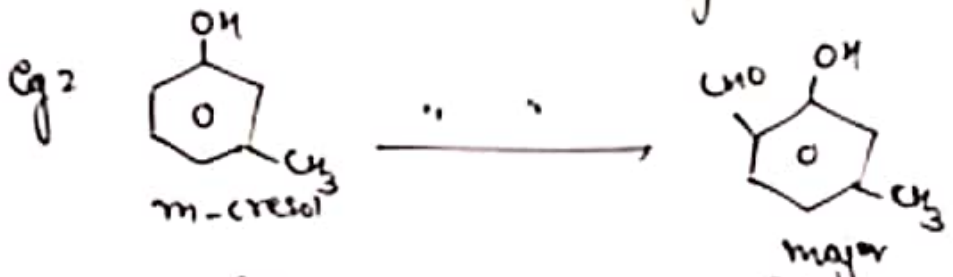
Scanned with CamScanner

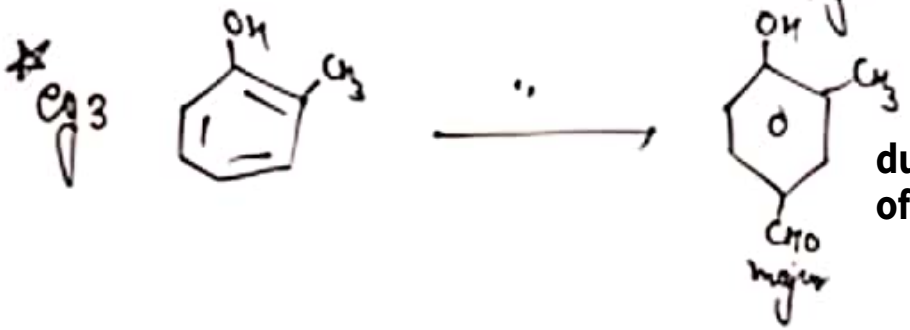
HROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

(6) Reimer - Tiemann's Reaction:-







due to high steric repulsion
of 3 adjacent groups, ortho product will not be major

Scanned with CamScanner

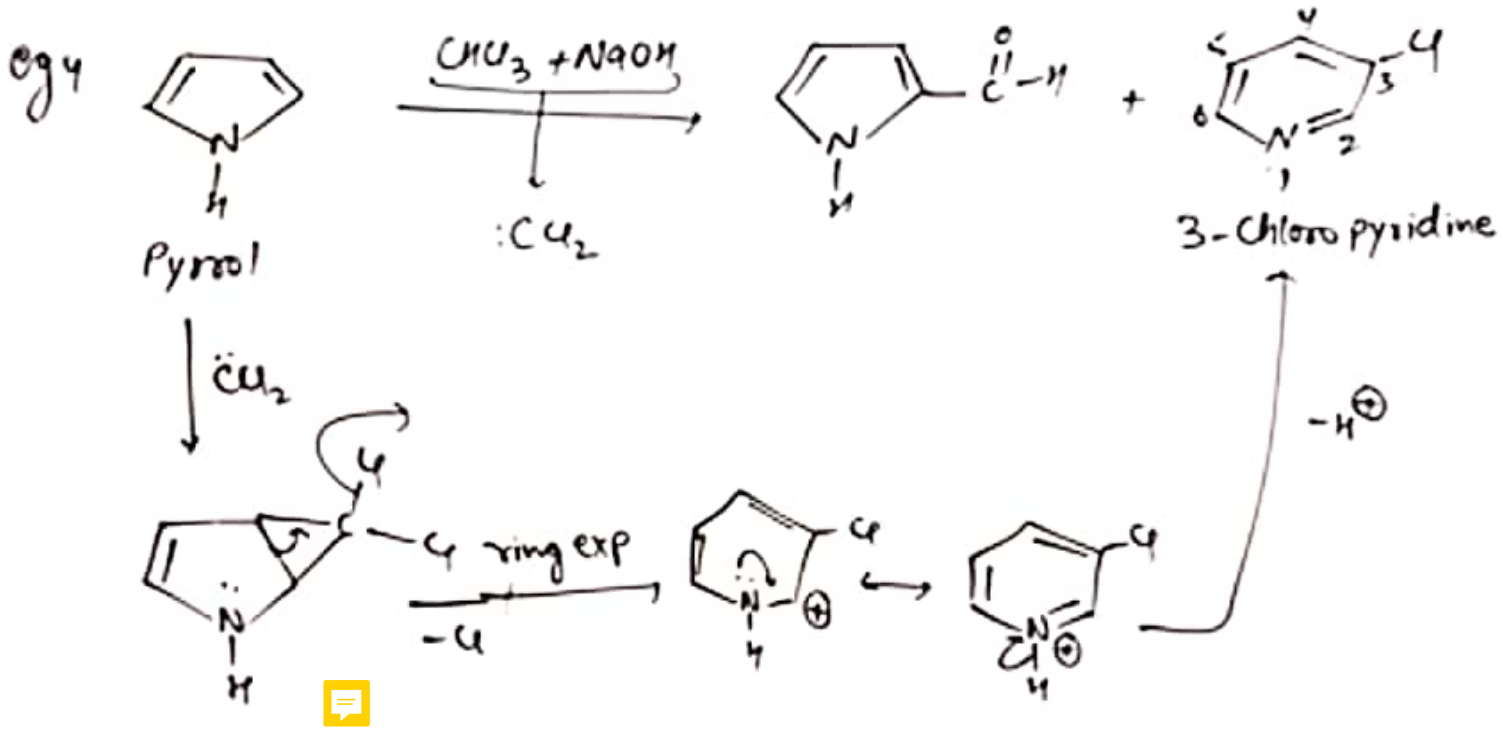
HROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

(6) Reimer - Tiemann's Reaction:-

numbering starts from the heteroatom in heterocyclic compounds





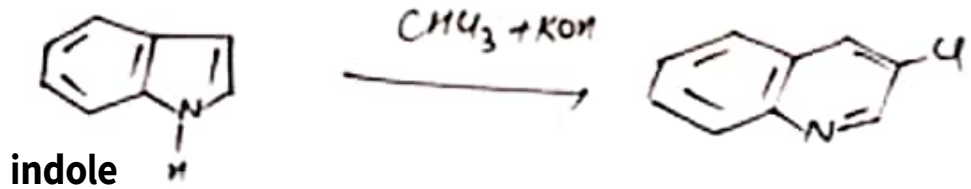
Scanned with CamScanner

HROMATIC COMPOUNDS

(I) PHENOL (Carbolic acid)

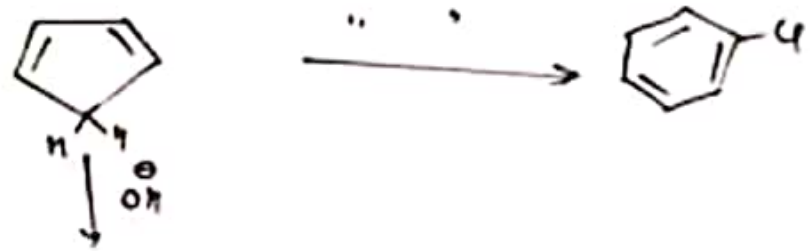
(6) Reimer - Tiemann's Reaction:-

eg 5



indole

eg 6





same as pyrrol

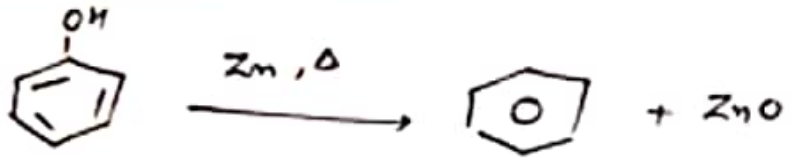
Scanned with CamScanner

HROMATIC COMPOUNDS

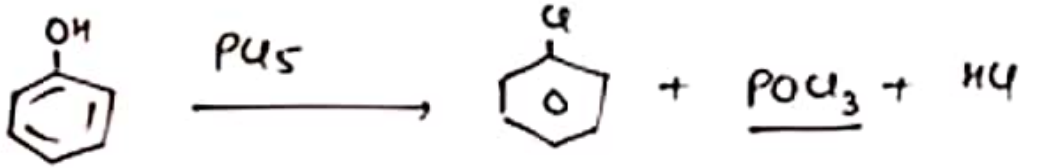
(V) PHENOL (carbolic acid)

→ Chemical Reactions of Phenol:-

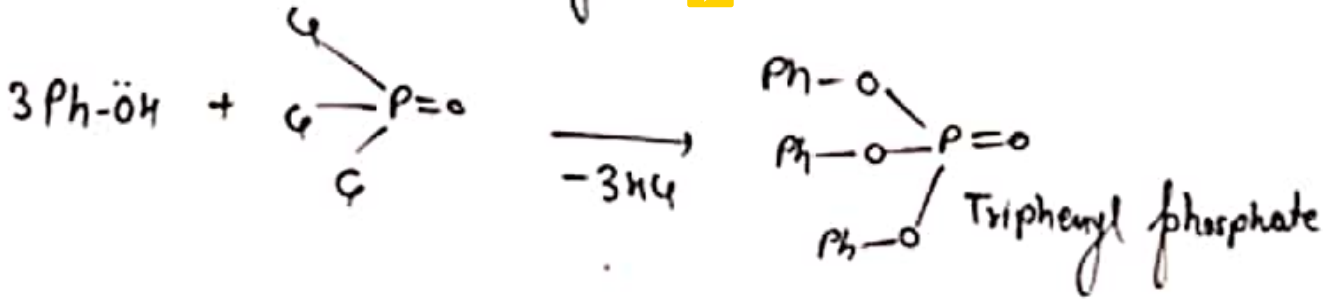
(7) Reaction with Zn:-



(8) Reaction PCl_5:-



Poor yield.



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HROMATIC COMPOUNDS

(V) PHENOL (carbolic acid)

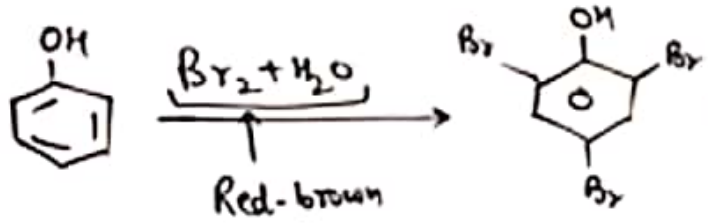
→ Chemical Reactions of Phenol:-

(9) Tests of Phenol

(9.A) Litmus test

Phenol can give blue litmus test

(9.B) $Br_2 + H_2O$ test



[>C=C< , -C≡C- can give Bromine water test)

and aniline

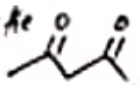
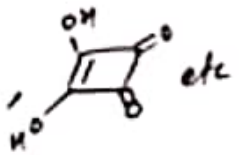
9.C Neutral $FeCl_3$ Test

$$Ph\text{-}OH + FeCl_3 \longrightarrow [Fe(OPh)_6]^{-3}$$

violet complex

- Other phenolic compounds → Blue, Green complex

- Enol can give +ve $FeCl_3$ test

like  ,  etc

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HROMATIC COMPOUNDS

(V) PHENOL (carbolic acid)

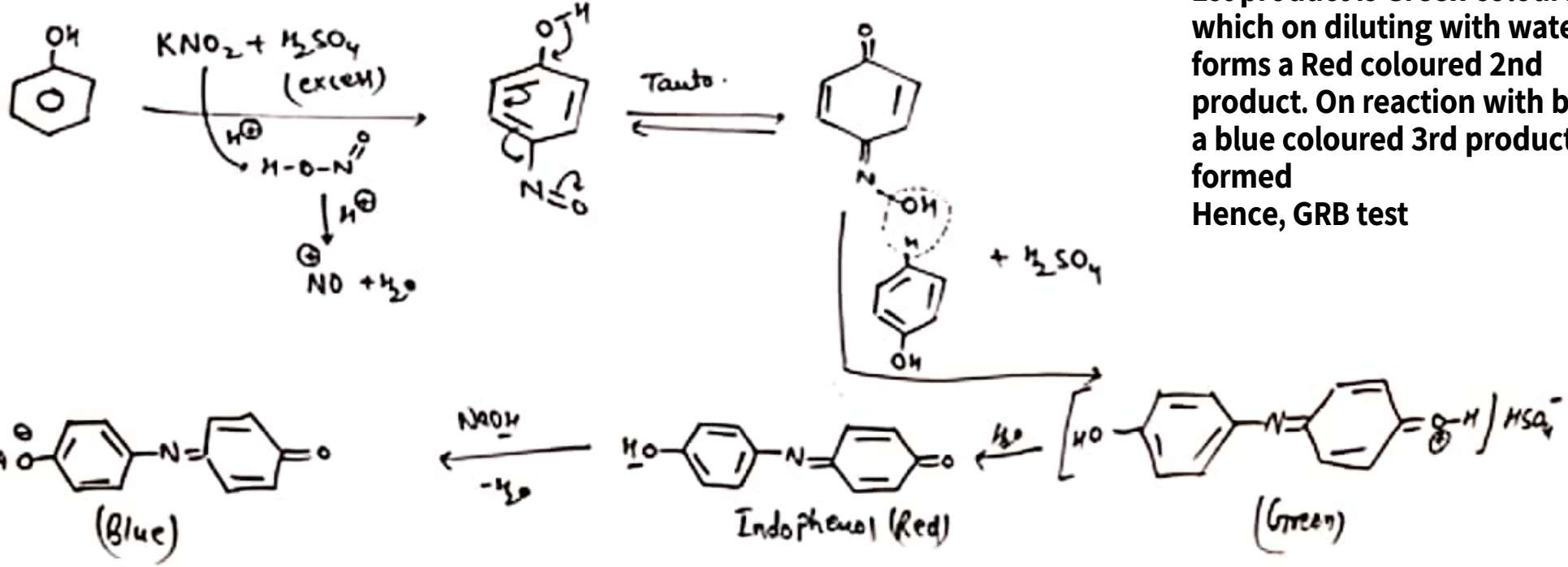
– Chemical Reactions of Phenol:-

(9) Tests of Phenols

9.D Libermann's Nitroso test:-

GRB Test

G → Green
R → Red
B → Blue



GRB Test since:

1st product is Green coloured, which on diluting with water forms a Red coloured 2nd product. On reaction with base, a blue coloured 3rd product is formed
Hence, GRB test

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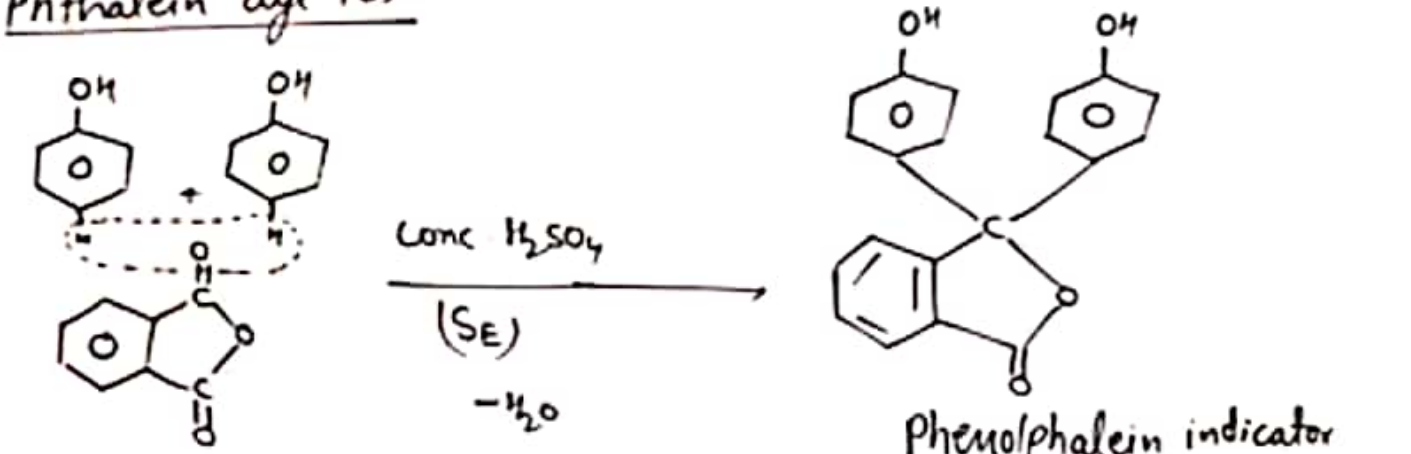
HROMATIC COMPOUNDS

(V) PHENOL (carbolic acid)

– Chemical Reactions of Phenol:-

(g) Tests of Phenols

(g E) Phthalein dye test



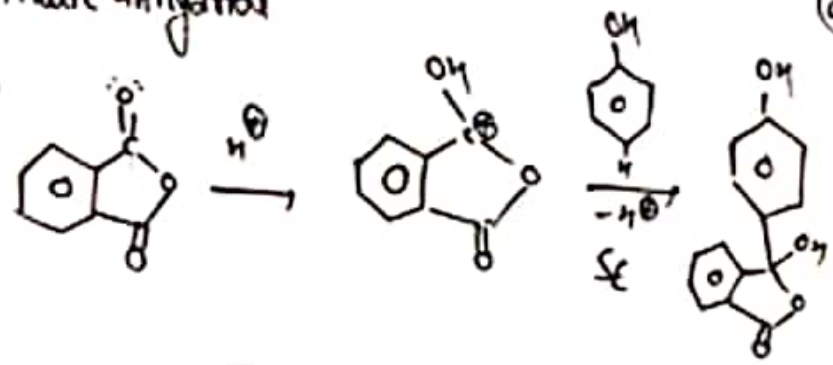
Phthalic anhydride

Phenolphalein indicator

(acid-base indicator)

pH range = 8.3 to 10

Hint mechanism



now, the acid again reacts with -OH undergoing EAS with a 2nd phenol molecule

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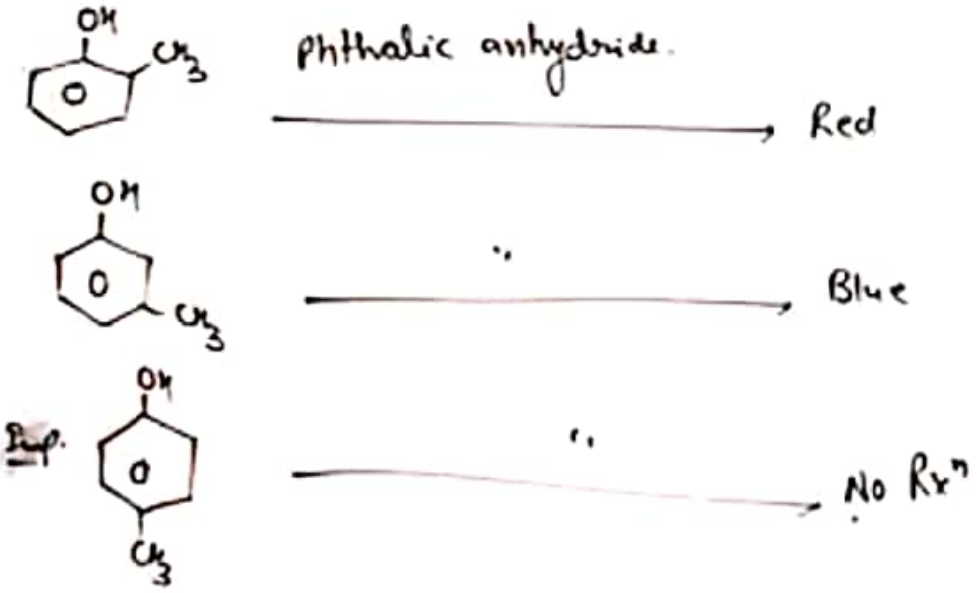
AROMATIC COMPOUNDS

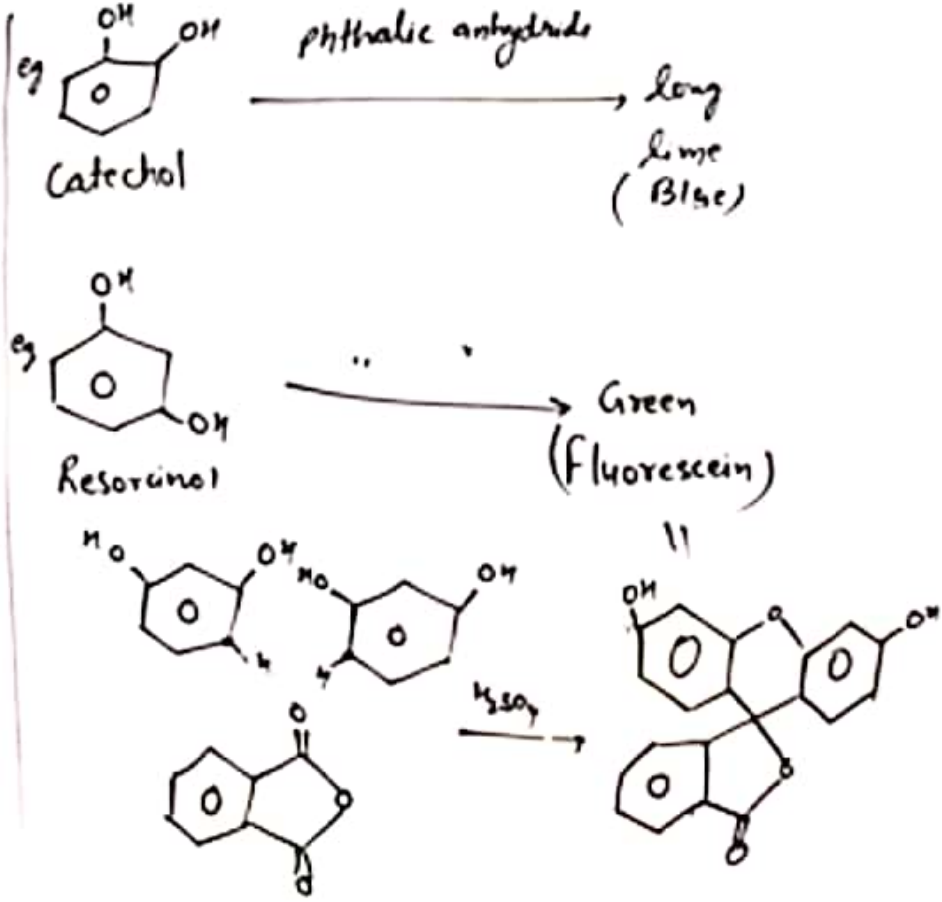
(V) PHENOL (carbolic acid)

– Chemical Reactions of Phenol:-

(9) Tests of Phenols

(9 E) Phthalein dye test





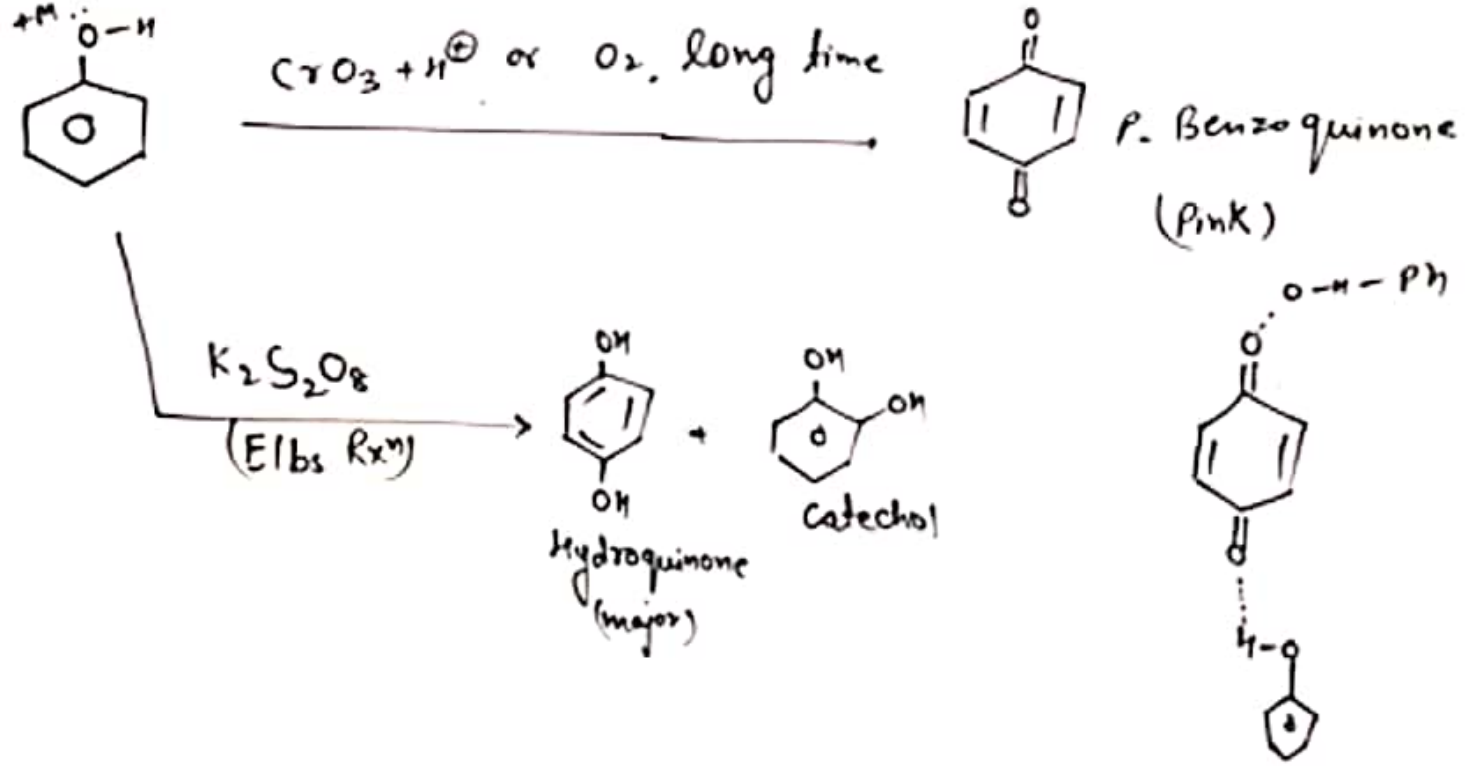
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AROMATIC COMPOUNDS

(V) PHENOL (carbolic acid)

– Chemical Reactions of Phenol:-

(10) Oxidation of Phenol



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AROMATIC COMPOUNDS

(V) PHENOL (carbolic acid)

→ Chemical Reactions of Phenol:-

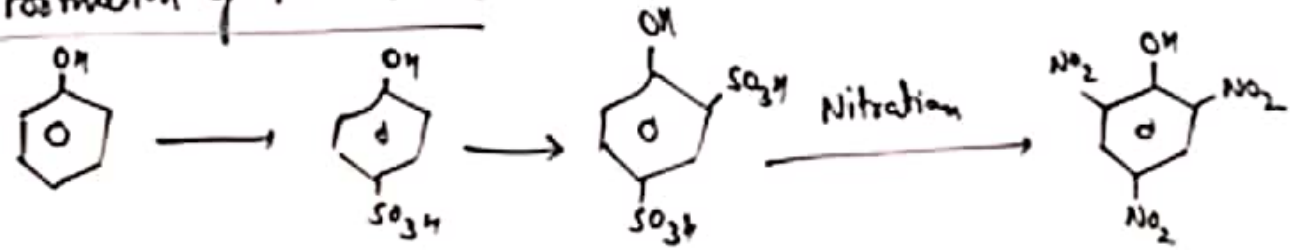
(11) Reaction of Phenol with diazonium salt (coupling Rxn)

(12) Nitration

(13) Sulphonation } ORM-II

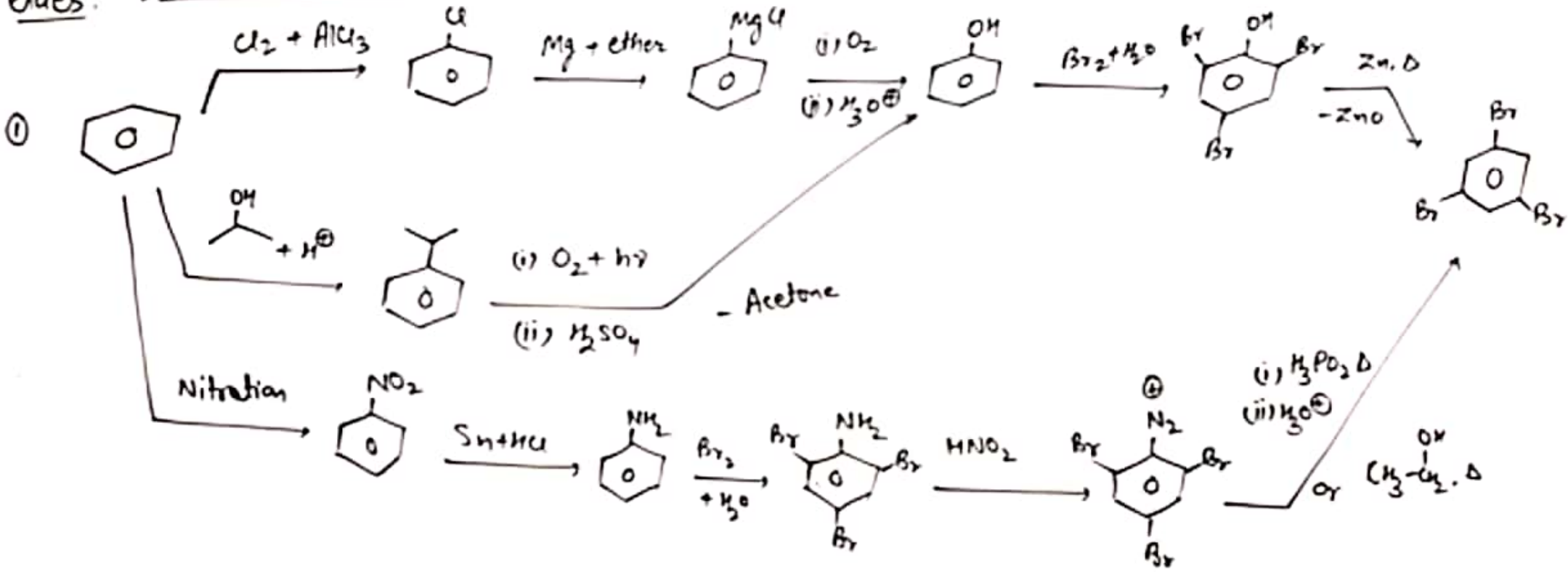
(14) Halogenation

→ Formation of Picric acid



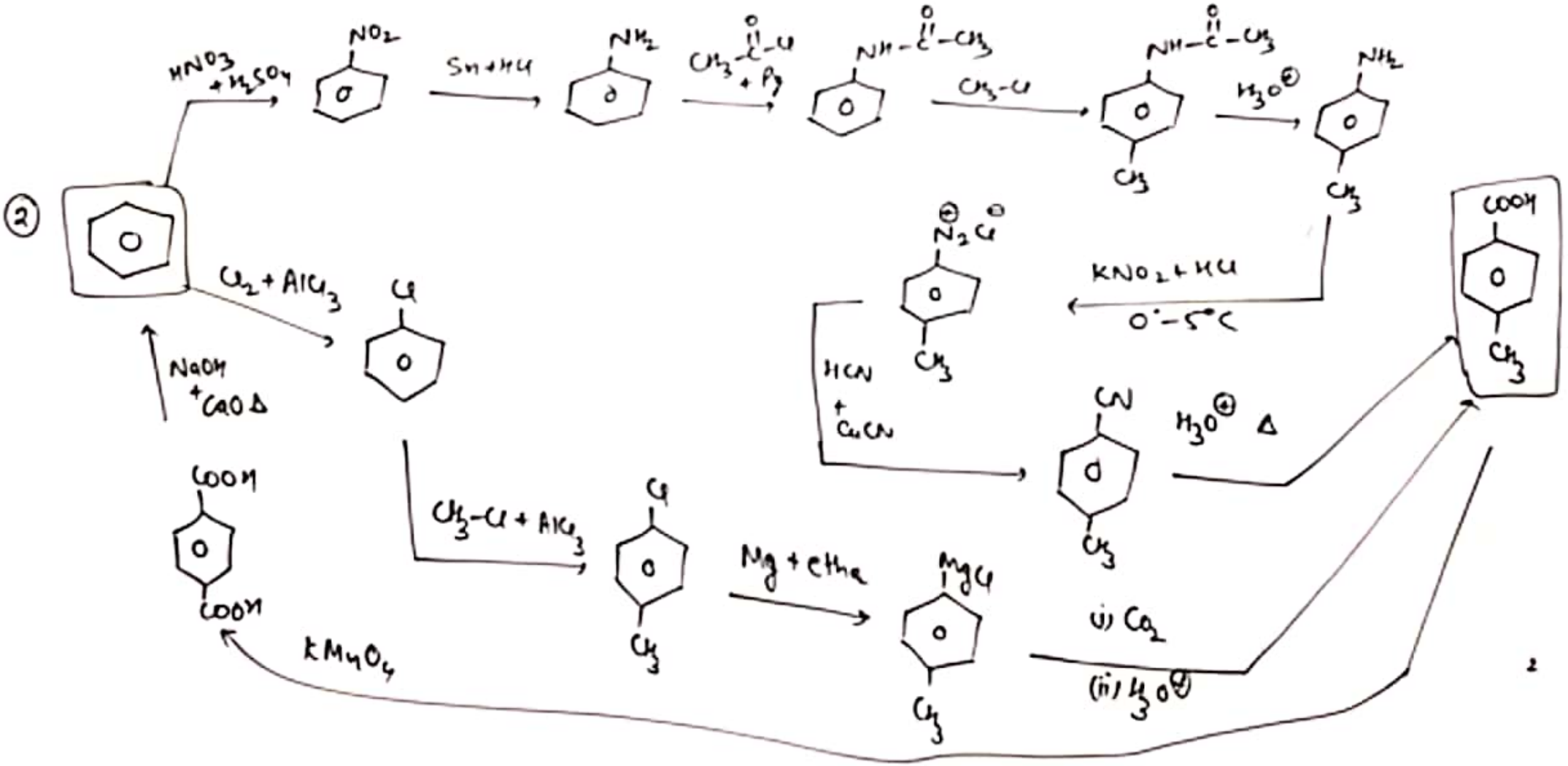
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Ques. Interconversions



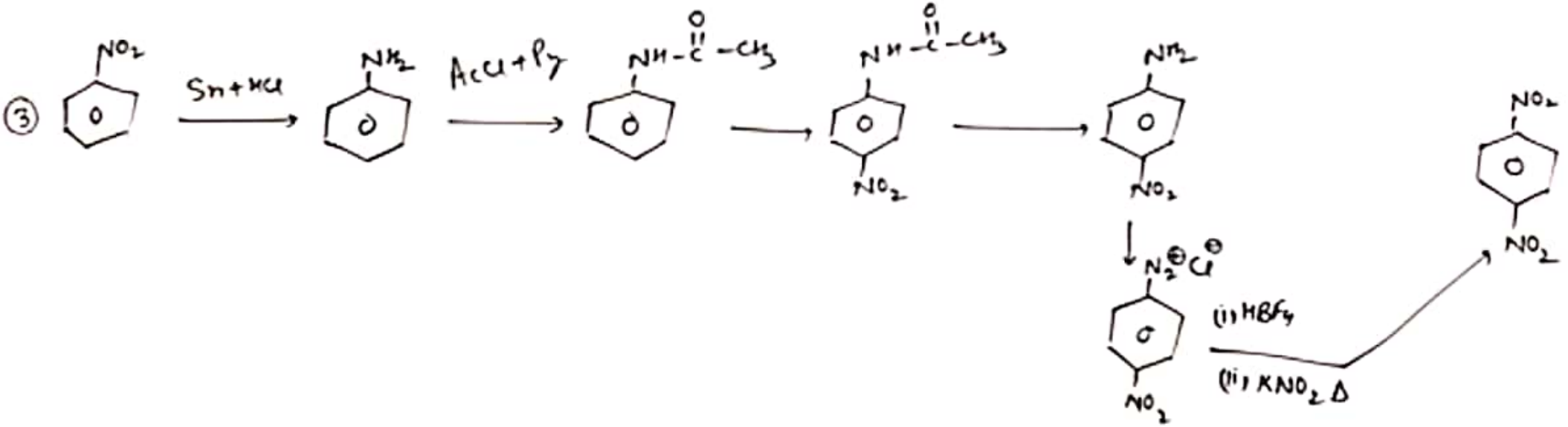
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Ques. Interconversions



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Ques. **Interconversions**

③ 

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